

Supplementary Materials

What Commentators Talk About: A Validated Thematic Taxonomy of Event-Grounded Sports Commentary

This document contains all supplementary materials for the paper. Eight supplements are included; each begins on a new page.

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S1 — Cluster Validation Cards (34 Clusters)

This supplement provides a per-cluster validation card for each of the 34 clusters in the released taxonomy. Cards are sorted by verdict band (strict-core first) and by cluster size within each band. Each card reports: domain, size, four-axis pass/fail indicators with raw scores, top R3 enriched terms, and three representative sentences. Axis criteria: Algorithm $\geq 4/7$ pipelines surviving (Jaccard ≥ 0.30); Bootstrap mean Jaccard ≥ 0.50 ; Held-out replication rate ≥ 0.50 ; Hierarchical Ward Jaccard ≥ 0.50 .

C6 — Grand Prix wins

Domain: Narrative & meta

Size: 465 sentences

Verdict band: Strict-core (4/4)

Four-axis validation:

	Axis	Score
✓	Algorithm	0.571 survival fraction
✓	Bootstrap	0.852 mean Jaccard
✓	Held-out	0.500 replication rate
✓	Hierarchical	0.686 mean Jaccard

Top R3 enriched terms: grand, grand prix, prix, wins, wins grand, lead grand, british, british grand, win, place grand, dutch, dutch grand, prix victory, lead, prix underway

Representative sentences:

1. *The Oscar goes to the top step of the podium, Oscar Piastri wins the Azerbaijan Grand Prix, and it will be a sequel to celebrate, the Australian is a grand prix winner for the second time.*
2. *The Mexican fans celebrating in that grandstand for Verstappen, who wins the Saudi Arabian Grand Prix with another drive of distinction.*
3. *To a man who's never won in Italy before, and who wins the Emilia Romagna Grand Prix this afternoon, Max Verstappen for the first time in his career takes victory in Italy.*

C0 — DRS-assisted battles

Domain: Race dynamics

Size: 205 sentences

Verdict band: Strict-core (4/4)

Four-axis validation:

	Axis	Score
✓	Algorithm	1.000 survival fraction
✓	Bootstrap	0.967 mean Jaccard
✓	Held-out	0.500 replication rate
✓	Hierarchical	0.820 mean Jaccard

Top R3 enriched terms: drs, got drs, drs zone, got, got, zone, open, drs enabled, enabled, going drs, bit drs, drs going, extra, drs open, drs time

Representative sentences:

1. *He's not got DRS, Fernando Alonso, to help his battle, Gasly has, giving him an extra speed advantage.*
2. *Whether he's going to be as close as Lando Norris was, let's see, he's got DRS in operation, he's getting that extra straight-line speed as they head down towards Stowe corner.*
3. *Now we've got DRS enabled, say goodbye to the Red Bull, that is absolutely outrageous, in another league when he used DRS.*

C31 — Race leader changes

Domain: Race dynamics

Size: 162 sentences

Verdict band: Strict-core (4/4)

Four-axis validation:

	Axis	Score
✓	Algorithm	0.571 survival fraction
✓	Bootstrap	0.635 mean Jaccard
✓	Held-out	0.500 replication rate
✓	Hierarchical	0.606 mean Jaccard

Top R3 enriched terms: race, lead race, lead, race leader, leader, takes lead, race lead, start race, started race, win race, going lead, race going, race control, lap race, race far

Representative sentences:

1. *In comes the race leader, Verstappen also being called in as well, so Verstappen absolutely now in contention for victory.*
2. *Well everyone has been lapped so far by the race leader down to Valtteri Bottas.*
3. *Race leader with a swap on, trying to take avoiding action and he clobbers the bollard out.*

C9 — Formula 1 milestones

Domain: Narrative & meta

Size: 134 sentences

Verdict band: Strict-core (4/4)

Four-axis validation:

	Axis	Score
✓	Algorithm	0.571 survival fraction
✓	Bootstrap	0.797 mean Jaccard
✓	Held-out	0.846 replication rate
✓	Hierarchical	0.671 mean Jaccard

Top R3 enriched terms: formula, formula world, time formula, podium formula, podium, formula career, world championship, world, championship, career, best formula, formula 2025, formula podium, 2025, time

Representative sentences:

1. *Welcome back to the Formula 1 World Championship in 2025.*
2. *A huge moment in the world championship, and it's Piastri, Verstappen, and on for his first podium in Formula 1 is Isack Hadjar.*
3. *But that is now half time officially in the Formula One world championship.*

C21 — Braking & lockups

Domain: Race dynamics

Size: 128 sentences

Verdict band: Strict-core (4/4)

Four-axis validation:

	Axis	Score
✓	Algorithm	0.571 survival fraction
✓	Bootstrap	0.579 mean Jaccard
✓	Held-out	0.600 replication rate
✓	Hierarchical	0.532 mean Jaccard

Top R3 enriched terms: braking, brakes, braking zone, zone, brake, late, late braking, brake pedal, pedal, lock, brakes just, way traffic, outside brakes, curb, brakes late

Representative sentences:

1. *He just brake-tested me, I got brake-tested like right before the braking zone, it looked like he actually went offline and ended up off the racing.*
2. *Laid on the brakes for Carlos Sainz, late-braking move and a position gained for Sainz who was distraught with being impeded when he had Isack Hadjar in the way yesterday.*
3. *Ocon is ahead going into the braking zone and Vettel locking up can't do anything to stop going into the rear.*

C16 — Safety car deployment

Domain: Regulatory

Size: 122 sentences

Verdict band: Strict-core (4/4)

Four-axis validation:

	Axis	Score
✓	Algorithm	1.000 survival fraction
✓	Bootstrap	0.689 mean Jaccard
✓	Held-out	0.562 replication rate
✓	Hierarchical	0.626 mean Jaccard

Top R3 enriched terms: safety, safety car, car, virtual, virtual safety, deployed, car deployed, car safety, ending, car ending, going safety, got safety, lapped cars, timing, car coming

Representative sentences:

1. *But it was a virtual safety car deployed, and Charles Leclerc comes into the pits.*
2. *The virtual safety car is now deployed, that was always coming, and everyone reduces their pace.*
3. *The virtual safety car has been deployed, so Leclerc's coming in, and Mercedes are waiting for George Russell.*

C1 — Time penalties

Domain: Regulatory

Size: 118 sentences

Verdict band: Strict-core (4/4)

Four-axis validation:

	Axis	Score
✓	Algorithm	1.000 survival fraction
✓	Bootstrap	0.966 mean Jaccard
✓	Held-out	0.926 replication rate
✓	Hierarchical	0.798 mean Jaccard

Top R3 enriched terms: penalty, time penalty, second penalty, second, second time, 10 second, causing collision, penalty causing, collision, causing, 10, time, penalty forcing, got second, forcing

Representative sentences:

1. *Alonso's been given a 10-second time penalty for causing a collision with Zhou Guanyu.*
2. *Charles Leclerc has been given a five second time penalty for leaving the track and gaining an advantage.*
3. *Kevin Magnussen has been given a 10-second time penalty for leaving the track and gaining an advantage.*

C8 — Race control flags

Domain: Regulatory

Size: 118 sentences

Verdict band: Strict-core (4/4)

Four-axis validation:

	Axis	Score
✓	Algorithm	0.857 survival fraction
✓	Bootstrap	0.772 mean Jaccard
✓	Held-out	0.605 replication rate
✓	Hierarchical	0.729 mean Jaccard

Top R3 enriched terms: flag, checkered, checkered flag, yellow, green, yellow flag, chequered, chequered flag, green flag, red flag, red, flag flies, sees checkered, sees, nation driver

Representative sentences:

1. *This is the final corner of the last lap, the Dutch driver looks up and sees a checkered flag first.*
2. *Charles Leclerc started second in the race today, he sees the checkered flag and for the first time other than when he's on pole he takes victory.*
3. *As for the first time he starts a race in Montreal on pole, and for the first time he sees the checkered flag.*

C28 — Corner battles

Domain: Race dynamics

Size: 82 sentences

Verdict band: Strict-core (4/4)

Four-axis validation:

	Axis	Score
✓	Algorithm	0.571 survival fraction
✓	Bootstrap	0.617 mean Jaccard
✓	Held-out	0.500 replication rate
✓	Hierarchical	0.675 mean Jaccard

Top R3 enriched terms: turn, run turn, abreast, way turn, run, heading turn, outside turn

Representative sentences:

1. *Lewis Hamilton's there alongside him, now he's on his back foot, now he's going to run wide through turn three.*
2. *Alonso is going to get the run down to turn number four, this is going to be super aggressive between the two of them.*
3. *Do we go three wide into turn number one?*

C22 — Wheel-to-wheel

Domain: Race dynamics

Size: 71 sentences

Verdict band: Strict-core (4/4)

Four-axis validation:

	Axis	Score
✓	Algorithm	0.857 survival fraction
✓	Bootstrap	0.711 mean Jaccard
✓	Held-out	0.556 replication rate
✓	Hierarchical	0.639 mean Jaccard

Top R3 enriched terms: wheel, wheel wheel, going wheel, wheel turn, wheel goes, goes wheel, went wheel, wheel action, wheel gets, wheels, went

Representative sentences:

1. *A Red Bull and a Mercedes going wheel to wheel into turn four, Perez stays within the white lines.*
2. *There's Sergio Perez and Charles Leclerc going wheel-to-wheel as they go round turn two.*
3. *Does look good if you're Pierre Gasly, that's for sure, he's going wheel to wheel with Lance Stroll into turn one.*

C10 — Chicane moves

Domain: Race dynamics

Size: 62 sentences

Verdict band: Strict-core (4/4)

Four-axis validation:

	Axis	Score
✓	Algorithm	0.857 survival fraction
✓	Bootstrap	0.853 mean Jaccard
✓	Held-out	1.000 replication rate
✓	Hierarchical	0.823 mean Jaccard

Top R3 enriched terms: chicane, second chicane, going chicane, goes chicane, braking, bold, chicane going, make chicane

Representative sentences:

1. *Now there's a lot of momentum, but again, no opportunity, so trying the different lines once more, but it's going to have to be a bold move into the second chicane.*
2. *And that's going to continue all the way down to the braking zone in the second chicane, great view from on board.*
3. *We're heading to the Roggia, and this is the second chicane on the racetrack on the second lap, and Stroll adjudged to be forced off the road.*

C20 — Wing & car damage

Domain: Strategy & technical

Size: 61 sentences

Verdict band: Strict-core (4/4)

Four-axis validation:

	Axis	Score
✓	Algorithm	0.714 survival fraction
✓	Bootstrap	0.626 mean Jaccard
✓	Held-out	0.625 replication rate
✓	Hierarchical	0.670 mean Jaccard

Top R3 enriched terms: wing, puncture, damage, damaged, got puncture, contact, bit contact, damage wing, damaged wing, rear, wing endplate, think, endplate, ripped, rear wing

Representative sentences:

1. *Yeah I think they touched actually, there was definitely a little jump from Piastri's car, and that would indicate to me, and it looks like the front wing endplate might be a little bit damaged on Lando.*
2. *Game on once again, this time it's the outside line, oh they make contact, he's got a puncture, he's damaged his front wing.*
3. *Yeah, Norris has a damaged front wing endplate where he's had that brush with Piastri.*

C11 — Team radio celebrations

Domain: Narrative & meta

Size: 55 sentences

Verdict band: Strict-core (4/4)

Four-axis validation:

	Axis	Score
✓	Algorithm	0.714 survival fraction
✓	Bootstrap	0.640 mean Jaccard
✓	Held-out	0.700 replication rate
✓	Hierarchical	0.769 mean Jaccard

Top R3 enriched terms: thank, job, mate, weekend, guys, amazing, fantastic, thanks, unbelievable, great job, thank guys, drive, good job, yeah, awesome

Representative sentences:

1. *Thank you very much guys, amazing strategy, we were really really sharp today, and nice pushing 72 laps here, I love it, great job guys.*
2. *What up Max, fantastic job, get her mate, hold on, yeah thanks guys that was a, I think a very good performance, very nice, good race, well executed, good pit stops.*
3. *Nice job guys, well done, excellent result for the team, fantastic, absolutely fantastic, really really amazing drive, that last thing was on fire, well done mate.*

C2 — Lights-out race starts

Domain: Race dynamics

Size: 54 sentences

Verdict band: Strict-core (4/4)

Four-axis validation:

	Axis	Score
✓	Algorithm	1.000 survival fraction
✓	Bootstrap	0.876 mean Jaccard
✓	Held-out	0.600 replication rate
✓	Hierarchical	0.703 mean Jaccard

Top R3 enriched terms: lights away, lights, away, away gets, away getting, start, lights lights, away ahead, start good, getting away, better start, gets away, away comes, reaction, better

Representative sentences:

1. *It's lights out and away we go and Max Verstappen gets the better start from Fernando Alonso.*
2. *It's lights out and away we go, Verstappen gets the better start, Perez and Lando Norris jockeying for position.*
3. *It's lights out and away we go and I think Verstappen got a slightly better start than Lewis Hamilton, and also Charles Leclerc a better start than Valtteri Bottas.*

C3 — Mirror checks defending

Domain: Race dynamics

Size: 47 sentences

Verdict band: Strict-core (4/4)

Four-axis validation:

	Axis	Score
✓	Algorithm	1.000 survival fraction
✓	Bootstrap	0.532 mean Jaccard
✓	Held-out	0.562 replication rate
✓	Hierarchical	0.728 mean Jaccard

Top R3 enriched terms: mirrors, mirror, checking, scarlet, checks, checking mirrors, checking left, hand mirror, left mirror, looking mirrors, left, checks mirrors, looking, checks mirror, driving mirrors

Representative sentences:

1. *There's a real opportunity now for Max Verstappen, and Charles Leclerc knows it, driving on the mirrors now, checking the left one, checking the right one, trying to keep the world champion behind.*
2. *P7, Tsunoda now checking his mirrors rather than looking ahead.*
3. *But Lando Norris was checking his mirrors a few corners ago, now he sees the crowd rise to applaud him as the winner.*

C27 — Overtake battles

Domain: Race dynamics

Size: 762 sentences

Verdict band: Robust (3/4)

Four-axis validation:

	Axis	Score
✓	Algorithm	0.571 survival fraction
✓	Bootstrap	0.688 mean Jaccard
✓	Held-out	0.556 replication rate
×	Hierarchical	0.339 mean Jaccard

Top R3 enriched terms: inside, door, corner, gets away, past, outside, open door, getaway, space, ahead, close, goes corner, leads corner, past outside, goes

Representative sentences:

1. *Max Verstappen on the inside has made a brilliant start but then goes off the track as Fernando Alonso closes the door on him.*
2. *Inside line for Lando Norris, roaring around the outside, will they make the corner?*
3. *Charles Leclerc tried to go past Lewis Hamilton, outside line to the inside line, got to be brave to make that work.*

C18 — Team dynamics

Domain: Narrative & meta

Size: 600 sentences

Verdict band: Robust (3/4)

Four-axis validation:

	Axis	Score
✓	Algorithm	0.857 survival fraction
✓	Bootstrap	0.709 mean Jaccard
✓	Held-out	1.000 replication rate
×	Hierarchical	0.365 mean Jaccard

Top R3 enriched terms: teammate, team, ahead teammate, ahead, past

Representative sentences:

1. *And there's the move for Daniel Ricciardo getting past his old teammate at turn five.*
2. *It's as you were ahead, here we go, the best chance to get past his old teammate.*
3. *This is Pierre Gasly on Yuki Tsunoda, big move and taking the place away from his old teammate and good friend, just out of the points.*

C15 — Tires & compounds

Domain: Strategy & technical

Size: 379 sentences

Verdict band: Robust (3/4)

Four-axis validation:

	Axis	Score
✓	Algorithm	0.857 survival fraction
✓	Bootstrap	0.827 mean Jaccard
×	Held-out	0.385 replication rate
✓	Hierarchical	0.575 mean Jaccard

Top R3 enriched terms: tires, tire, tyres, hard, medium, compound, tyre, soft, fresher, compound tire, fresher tires, hard compound, medium tires, hard tires, intermediate

Representative sentences:

1. *So on the medium tires that are fresher as Valtteri Bottas is hard compound tire, and with DRS as well he gains on the Mercedes.*
2. *Hard tires going on to Lewis Hamilton's Mercedes, they've not got any new sets of mediums.*
3. *Yes yes everyone's kept a set of new hard compound tires just for the middle phase of this race.*

C14 — Pit stops

Domain: Strategy & technical

Size: 185 sentences

Verdict band: Robust (3/4)

Four-axis validation:

	Axis	Score
✓	Algorithm	1.000 survival fraction
✓	Bootstrap	0.776 mean Jaccard
✓	Held-out	0.667 replication rate
×	Hierarchical	0.447 mean Jaccard

Top R3 enriched terms: pit, pit stop, lane, pit lane, stop, pits, coming pit, comes pit, come pits, comes, coming, come pit, pitted, stops, slow

Representative sentences:

1. *And my word, Lance Stroll is making a pit stop, something is happening in the pit lane in front of us.*
2. *He is a contender to win this one, as Lando Norris comes in to make his pit stop, leaves into the fast lane he goes, 2.3 seconds and Lando Norris lights up the rear tires, goes back to work.*
3. *Hamilton comes in, Verstappen and Perez stay out, Hamilton it's a slow stop again but he had to wait for the other car that was coming down the pit lane.*

C13 — Championship & title

Domain: Narrative & meta

Size: 158 sentences

Verdict band: Robust (3/4)

Four-axis validation:

	Axis	Score
✓	Algorithm	0.714 survival fraction
✓	Bootstrap	0.693 mean Jaccard
✓	Held-out	0.600 replication rate
×	Hierarchical	0.492 mean Jaccard

Top R3 enriched terms: championship, world, world championship, champion, world champion, title, constructors, win, time world, championship leader, trophy, 2025, points, leader, 2024

Representative sentences:

1. *The world championship leader up against the two-time world champion highly regarded Fernando Alonso.*
2. *The world championship leader Oscar Piastri didn't finish, but he still leads by 25 points.*
3. *As we go on to the penultimate tour, Piastri who leads the championship by nine points trying to score win number six of the World Championship.*

C12 — Podium finishes

Domain: Narrative & meta

Size: 139 sentences

Verdict band: Robust (3/4)

Four-axis validation:

	Axis	Score
✓	Algorithm	1.000 survival fraction
✓	Bootstrap	0.826 mean Jaccard
×	Held-out	0.462 replication rate
✓	Hierarchical	0.767 mean Jaccard

Top R3 enriched terms: podium, podium places, home, place podium, comes home, podium place, completes, places, finish, completes podium, podium finish, complete podium, step, step podium, stand podium

Representative sentences:

1. *Max Verstappen completes the podium with third place, and Charles Leclerc in his own race comes home to finish, for the first time.*
2. *Sergio Perez provided great support to his teammate and to his team to come home and take the final podium place as well.*
3. *As Charles Leclerc comes home for his first podium finish of the season.*

C30 — Lap pace & timing

Domain: Strategy & technical

Size: 136 sentences

Verdict band: Robust (3/4)

Four-axis validation:

	Axis	Score
×	Algorithm	0.429 survival fraction
✓	Bootstrap	0.697 mean Jaccard
✓	Held-out	0.900 replication rate
✓	Hierarchical	0.534 mean Jaccard

Top R3 enriched terms: lap, fastest lap, fastest, opening, previous lap, opening lap, previous, lap lap, turn lap, lap goes, lap going, lap time, quicker, end lap, middle

Representative sentences:

1. *Perez in clear air, finally away from the back markers, 1:19.468 new fastest lap, gap down to six seconds.*
2. *And he might just be slowing down a bit as he crosses the line, he did slow down, he did abort the fastest lap attempt.*
3. *There's the championship points at the moment, Hamilton picking up 19 because of that fastest lap.*

C33 — Field position changes

Domain: Race dynamics

Size: 122 sentences

Verdict band: Robust (3/4)

Four-axis validation:

	Axis	Score
✓	Algorithm	0.571 survival fraction
✓	Bootstrap	0.510 mean Jaccard
✓	Held-out	0.619 replication rate
×	Hierarchical	0.392 mean Jaccard

Top R3 enriched terms: seventh, place, sixth, fifth, sixth place, eighth, seventh place, 11th, position, fifth place, sixth position, 10th, ninth, 12th, seventh position

Representative sentences:

1. *Norris is up to ninth, P10 for Lewis Hamilton, Russell versus Albon, this is for sixth position, or it was, because Albon has just driven around the side of his good friend and taken sixth place.*
2. *He moves up to sixth position, Perez down to seventh.*
3. *Oscar Piastri still in fifth position, no change for anyone until you look all the way back down to Lewis Hamilton who's up to seventh position, gaining a couple of places, as well as Hülkenberg.*

C19 — Race lead holding

Domain: Race dynamics

Size: 115 sentences

Verdict band: Robust (3/4)

Four-axis validation:

	Axis	Score
✓	Algorithm	0.857 survival fraction
✓	Bootstrap	0.694 mean Jaccard
×	Held-out	0.300 replication rate
✓	Hierarchical	0.620 mean Jaccard

Top R3 enriched terms: lead, leads, leads second, lead second, maintains lead, holds lead, lead comes, holds, leads comes, maintains, leads inside, leads leads, takes lead, maintain lead

Representative sentences:

1. *Lando Norris leads, yeah that was a great start from him, I think there was some contact we saw a little puff of smoke but no visible damage, and uh yeah what a great great start for Lando Norris.*
2. *Hamilton leads from Verstappen, then comes Sergio Perez and then the Ferrari of Charles Leclerc.*
3. *Leclerc leads Verstappen on the inside, Sainz on the outside, Verstappen stays ahead going into turn one.*

C5 — Weather conditions

Domain: Strategy & technical

Size: 43 sentences

Verdict band: Robust (3/4)

Four-axis validation:

	Axis	Score
✓	Algorithm	1.000 survival fraction
×	Bootstrap	0.075 mean Jaccard
✓	Held-out	0.667 replication rate
✓	Hierarchical	0.716 mean Jaccard

Top R3 enriched terms: rain, rain coming, wet, minutes, rain falling, falling, light, expected, track, extraordinary, class, lasting, temperature, 10 minutes, 30

Representative sentences:

1. *Coming out of turn three, the light rain falling, and the seven-time world champion has found the barriers.*
2. *Rain expected in seven laps lasting for 10 minutes, and if it's heavy rain, 10 minutes of it is going to really change things.*
3. *Yeah it is just that light rain, it's still falling, but it's not enough now, and I think the teams are getting more confidence that it's going to pass without significantly dampening the track.*

C4 — Race start launch

Domain: Race dynamics

Size: 42 sentences

Verdict band: Robust (3/4)

Four-axis validation:

	Axis	Score
✓	Algorithm	1.000 survival fraction
×	Bootstrap	0.358 mean Jaccard
✓	Held-out	0.600 replication rate
✓	Hierarchical	0.861 mean Jaccard

Top R3 enriched terms: launch, launch line, good launch, line, good, got good, got launch, gets launch, chops, pole sitter, sitter, chop, got, initial launch, great launch

Representative sentences:

1. *A good launch off the line for Lando Norris, sees him chop across Charles Leclerc, but Charles got a great amount of momentum but was blocked.*
2. *Good launch off the line for our pole-sitter Oscar Piastri, but also George Russell as well who's alongside Max Verstappen.*
3. *Verstappen's got a good launch, he's already up to second, I reckon that's the race lead in the inside line.*

C26 — Defending & attacking

Domain: Race dynamics

Size: 110 sentences

Verdict band: Moderate (2/4)

Four-axis validation:

	Axis	Score
✓	Algorithm	0.714 survival fraction
✓	Bootstrap	0.664 mean Jaccard
×	Held-out	0.333 replication rate
×	Hierarchical	0.481 mean Jaccard

Top R3 enriched terms: fight, defend, battle, defensive, going fight, try defend, defended, try fight, defense, fights, going

Representative sentences:

1. *Fernando Alonso knew what was coming, he fights his way by, good battle between these two, now resolved in favor of Alonso.*
2. *And now Stroll up against Ocon is the battle we're looking at, and Ocon's going to fight away by, now these two get on pretty well, they still get on brilliantly.*
3. *Perez fights it with one hand tied behind his back a little bit, he wasn't going to put up too much defense.*

C23 — Circuit action

Domain: Race dynamics

Size: 72 sentences

Verdict band: Moderate (2/4)

Four-axis validation:

	Axis	Score
×	Algorithm	0.143 survival fraction
✓	Bootstrap	0.700 mean Jaccard
×	Held-out	0.400 replication rate
✓	Hierarchical	0.585 mean Jaccard

Top R3 enriched terms: racetrack, corner racetrack, turn racetrack, turns racetrack

Representative sentences:

1. *Oliver Bearman, Hamilton clears Max Verstappen, no fireworks compared to what we've seen before from these two, notably around this racetrack in 2021.*
2. *Piastrri on a different part of the racetrack, it is dirty and dusty off the racing line, driving on his mirrors now the rookie Australian.*
3. *And Piastrri going very brave to a dirty dusty part of the racetrack to make the move into turn 23.*

C17 — Gap & pace

Domain: Strategy & technical

Size: 67 sentences

Verdict band: Moderate (2/4)

Four-axis validation:

	Axis	Score
✓	Algorithm	1.000 survival fraction
×	Bootstrap	0.461 mean Jaccard
×	Held-out	0.333 replication rate
✓	Hierarchical	0.602 mean Jaccard

Top R3 enriched terms: seconds, gap, tenths, hour, miles hour, tenths second, gap seconds, miles, half, extra, 20, pace, 10 seconds, second, 10

Representative sentences:

1. *The gap at the front, yeah, 5 and a half seconds, Lando doing his best lap of the race, but Oscar's holding that pace firm, it's tight, it's tight.*
2. *And don't forget the gap was about one and a half seconds, point seven it was yeah, between Verstappen and Hamilton.*
3. *We certainly are, 20.6 for Hamilton, that's his pace, 1.2 seconds taking out of him, that gap down to six seconds.*

C7 — Race start lights

Domain: Race dynamics

Size: 36 sentences

Verdict band: Moderate (2/4)

Four-axis validation:

	Axis	Score
×	Algorithm	0.429 survival fraction
×	Bootstrap	0.041 mean Jaccard
✓	Held-out	0.636 replication rate
✓	Hierarchical	0.864 mean Jaccard

Top R3 enriched terms: lights, red lights, prix underway, underway, grand, grand prix, prix, red, lights time, lights away, green flag, begins, racing lights, green, away

Representative sentences:

1. *The green flag at the back tells us we go racing in five lights time and the Austrian Grand Prix is underway.*
2. *One, two, three, four, five red lights in Budapest, and the Hungarian Grand Prix is underway.*
3. *Two, three, four, five red lights in Budapest, and the Hungarian Grand Prix is underway.*

C29 — Corner specific moves

Domain: Race dynamics

Size: 60 sentences

Verdict band: Weak (1/4)

Four-axis validation:

	Axis	Score
×	Algorithm	0.143 survival fraction
×	Bootstrap	0.445 mean Jaccard
×	Held-out	0.375 replication rate
✓	Hierarchical	0.717 mean Jaccard

Top R3 enriched terms: turn, turn 11, 11, turn 10, turn seven, 13, turn 12, 12, inside turn, seven, heading turn, heading, coming turn, turn 13, 10

Representative sentences:

1. *We'll send it to turn seven, high speed stuff all the time.*
2. *But let's have a look once again at this part, because now we're on the short burst up towards turn seven.*
3. *Nico Hülkenberg following into turn seven and is, no, I wondered if that was the moment.*

C25 — Driver & team references

Domain: Narrative & meta

Size: 69 sentences

Verdict band: Fragile (0/4)

Four-axis validation:

	Axis	Score
×	Algorithm	0.143 survival fraction
×	Bootstrap	0.472 mean Jaccard
×	Held-out	0.455 replication rate
×	Hierarchical	0.481 mean Jaccard

Top R3 enriched terms: driver, nation driver, nation, drivers, driver getting, board nation, driver trying, contract

Representative sentences:

1. *McLaren started this weekend by announcing they've got both drivers signed up long term, Oscar Piastri has a bumper contract, and he has the race leader close ahead of him.*
2. *So Yuki doesn't love that idea, we're talking about working with a senior team, they got to work with their own drivers at the moment.*
3. *And in we come at the same time, McLaren and Mercedes elect to pit their drivers, Antonelli and Piastri and Hamilton all in on the same lap.*

C24 — Inter-car distance

Domain: Strategy & technical

Size: 60 sentences

Verdict band: Fragile (0/4)

Four-axis validation:

	Axis	Score
×	Algorithm	0.143 survival fraction
×	Bootstrap	0.411 mean Jaccard
×	Held-out	0.417 replication rate
×	Hierarchical	0.475 mean Jaccard

Top R3 enriched terms: car, cars, car length, car lengths, length, lengths, couple car, width, got car, length ahead, car ahead, car width, flat floor, floor, flat

Representative sentences:

1. *And this is the closest yet Perez has been, yes it is, and he's only got a couple of car lengths.*
2. *And he's got a quarter of a car length in front now, they come out of turn two and he's got a couple of car lengths in front.*
3. *Alonso makes his way out of the first corner and he's only a couple of car lengths behind Leclerc.*

C32 — Front position changes

Domain: Race dynamics

Size: 45 sentences

Verdict band: Fragile (0/4)

Four-axis validation:

	Axis	Score
×	Algorithm	0.000 survival fraction
×	Bootstrap	0.241 mean Jaccard
×	Held-out	replication rate
×	Hierarchical	0.407 mean Jaccard

Top R3 enriched terms: fourth, place, position, fourth place, second, second place, couple positions

Representative sentences:

1. *Up to fourth place goes Pierre Gasly, down a couple of positions goes Kevin Magnussen.*
2. *Charles Leclerc comes out in fourth place.*
3. *Fourth place now for Carlos Sainz.*

S2 — Human-Rater Survey Protocol and Instrument

This supplement documents the design and administration of the human-rater validation study. It covers the intake question, task structure, sentence presentation, distractor design, domain definitions, and the reliability analysis framework. Actual survey responses are released separately as de-identified CSV files.

1. Overview

The human-rater study evaluates the 34-cluster taxonomy through two complementary evidence sources: **inter-rater reliability** (IRR) and **category recognition**. Nine raters with varying self-reported F1 familiarity (3 at level 2, 3 at level 3, 2 at level 4, 1 at level 5) completed a structured survey. The survey instrument was delivered via Google Forms and comprised one intake question followed by 34 cluster sections (one per cluster), each containing five tasks. Total mandatory questions: $1 + 34 \times 5 = 171$.

Parameter	Value
Raters (N)	9
Clusters evaluated	34
Tasks per cluster	5
Estimated time per rater	60–75 minutes
Platform	Google Forms
Total mandatory questions	171

2. Intake Question

Before the cluster sections, each rater answered:

“How familiar are you with Formula 1 broadcasts?”

Responses used a five-point ordinal scale:

Level	Wording
1	No prior experience with F1 broadcasts.
2	Occasional exposure; familiar with basic concepts (overtakes, pit stops).
3	Occasional viewer; recognise some drivers and teams.
4	Regular viewer; follow the current season actively.
5	Long-term follower; familiar with technical and strategic terminology.

3. Sentence Presentation

Each cluster section presented five sentences: four sampled from the target cluster and one out-of-cluster intruder. Sentences were drawn from the highlights corpus and shown in **original wording** (not entity-masked), preceded by a disclosure note that the underlying clustering analysis was performed on entity-masked text. Sentence order was randomised once at generation time (RNG seed = 42); the intruder position varied across sections (distribution across positions *a–e*: 6, 6, 8, 6, 8).

4. Task Order (per cluster section)

Tasks were presented in the following fixed order to enforce a **blind free-label** (Task 2 precedes Task 3 so that candidate labels do not prime the rater’s own description):

#	Task	Description	Chance
1	Intrusion detection	Pick the sentence that does not belong (5 choices + “All seem to belong”)	$\frac{1}{6} \approx 16.7\%$
2	Free label	Describe the theme in 1–4 words (free text; collected before label choices are shown)	—
3	Label selection	Choose the best label from 3 options + “None of these fit”	$\frac{1}{4} = 25.0\%$
4	Domain assignment	Choose the best broad domain from 4 options + “Doesn’t fit any”	$\frac{1}{5} = 20.0\%$
5	Importance rating	Rate the topic’s importance for understanding a race on a 1–5 Likert scale	—

Blindness enforcement. Each cluster section was split across two pages with a mandatory page break between Task 2 and Task 3. Page 1 showed the five sentences, Task 1, and Task 2; Page 2 re-displayed the same five sentences and then Tasks 3–5. This prevented candidate labels from being visible while the rater was filling in their free-label response.

5. Domain Options (Task 4)

Domain	Description shown to raters
Race dynamics	What happens on track (overtakes, defending, corners, starts)
Narrative & meta	Storylines, outcomes, records, championships, podiums
Strategy & technical	Race conditions and decisions (tyres, pit, pace, gaps, weather, damage)
Regulatory	Race control and rules (penalties, flags, safety car)
Doesn’t fit any	—

6. Importance Scale (Task 5)

1–5 Likert with anchors: 1 = *minor background element*; 5 = *central to understanding the race narrative*. The scale is ordinal; mean values per cluster are reported descriptively.

7. Distractor Design (Task 3 — label selection)

For each cluster, the three label options were: (1) the researcher-assigned thematic label; (2) the label of the nearest same-domain cluster by centroid cosine; (3) the label of the second-nearest same-domain cluster. Where the same-domain pool was too small or thematically too narrow (≈ 6 of 34 sections), distractors fell back to the nearest other-domain clusters. For C2 and C7 (both start-related), the near-synonym distractor was swapped to a non-start same-domain cluster to avoid near-identical choices. The position of the researcher label ($a/b/c$) was shuffled across sections; the ground-truth `ours_letter` field records the actual position used.

8. Intruder Selection (Task 1)

The intruder sentence was drawn from a different broad domain where possible (cross-domain intruder). In 5 of 34 sections where cross-domain candidates were exhausted, the intruder was drawn from a same-domain cluster instead. The ground-truth `intruder_letter` records the shuffled position of the intruder sentence.

9. Reliability Analysis Plan

IRR was computed per task across all nine raters:

- Krippendorff’s α with bootstrap 95% confidence intervals (Hayes & Krippendorff, 2007)
- Gwet’s AC1 as a prevalence-corrected complement (Gwet, 2008)

- Interpretive thresholds: $\alpha \geq 0.667$ tentative agreement, $\alpha \geq 0.800$ high (Krippendorff, 2004)

Tasks 1 and 4 reached the tentative-agreement threshold ($\alpha = 0.671$ and $\alpha = 0.687$ respectively); Task 3 ($\alpha = 0.516$) is treated as descriptive evidence only.

10. Data Availability

The full survey instrument (170 sentences across 34 sections) is not redistributed because of broadcast copyright. One illustrative section is included in the released code repository. De-identified response data (letter choices and importance ratings, no free text) and the analysis scripts are released openly. The full instrument is available upon reasonable request for non-commercial research use.

S3 — Algorithm-Axis Per-Cluster Survival Data

This supplement reports the full per-cluster Algorithm-axis survival data for the seven pre-specified alternative pipelines. Each cell shows the best Hungarian-Jaccard score between the primary cluster and the matched cluster in the alternative pipeline. Green cells (≥ 0.30) count as surviving. Row background colour indicates verdict band. The first section reproduces Table 3 from the main paper for reference.

Pipeline overview (Table 3 from main paper)

Pipeline	DBCV $\uparrow$	Silhouette $\uparrow$	n clusters	Noise %
MiniLM+UMAP+HDBSCAN (primary)	0.348	0.662	34	33.5
MPNet+UMAP+HDBSCAN	0.113	0.308	33	35.1
BGE+UMAP+HDBSCAN	0.237	0.611	35	45.2
GTE+UMAP+HDBSCAN	0.253	0.556	32	41.9
MiniLM+UMAP+KMeans	−0.375	0.440	34	0.0
MiniLM+UMAP+Ward	−0.341	0.420	34	0.0
MiniLM+UMAP+OPTICS	0.234	0.780	36	68.0
MiniLM+UMAP+Leiden	−0.209	0.401	39	0.0

Per-cluster survival matrix

Table 1: Per-cluster Jaccard survival scores across seven alternative pipelines. A cell is green if Jaccard ≥ 0.30 (the Algorithm-axis pass threshold). The rightmost column (N) gives the total survival count.

C#	Label	Size	MPNet	BGE	GTE	OPTICS	KMeans	Ward	Leiden	N
0	DRS-assisted battles	205	0.791	0.741	0.828	1.000	1.000	1.000	1.000	7
1	Time penalties	118	0.442	0.756	0.840	0.483	1.000	1.000	1.000	7
2	Lights-out race starts	54	0.724	0.721	0.767	1.000	1.000	1.000	1.000	7
3	Mirror checks defending	47	0.592	0.667	0.653	1.000	1.000	1.000	1.000	7
4	Race start launch	42	0.653	0.434	0.479	1.000	1.000	0.494	1.000	7
5	Weather conditions	43	0.580	0.623	0.592	1.000	0.524	0.426	1.000	7
6	Grand Prix wins	465	0.269	0.228	0.241	0.329	0.989	0.981	0.989	4
7	Race start lights	36	0.225	0.650	0.234	0.878	0.224	0.222	0.818	3
8	Race control flags	118	0.721	0.019	0.361	0.576	0.733	0.728	0.983	6
9	Formula 1 milestones	134	0.086	0.097	0.128	0.440	0.924	0.957	0.937	4
10	Chicane moves	62	0.038	0.358	0.587	1.000	0.969	1.000	0.984	6
11	Team radio celebrations	55	0.617	0.571	0.613	0.846	0.165	0.166	0.885	5
12	Podium finishes	139	0.527	0.556	0.518	0.403	0.891	0.885	0.946	7
13	Championship & title	158	0.103	0.436	0.250	0.316	0.469	0.476	0.558	5
14	Pit stops	185	0.472	0.515	0.540	0.341	0.907	0.907	0.881	7
15	Tires & compounds	379	0.687	0.675	0.733	0.198	0.959	0.942	0.959	6
16	Safety car deployment	122	0.691	0.645	0.604	0.680	0.931	0.946	0.693	7
17	Gap & pace	67	0.319	0.395	0.432	0.774	0.489	0.660	0.606	7
18	Team dynamics	600	0.410	0.561	0.570	0.000	0.628	0.824	0.792	6
19	Race lead holding	115	0.045	0.306	0.303	0.935	0.426	0.710	0.444	6

continued...

C#	Label	Size	<i>MPNet</i>	<i>BGE</i>	<i>GTE</i>	<i>OPTICS</i>	<i>KMeans</i>	<i>Ward</i>	<i>Leiden</i>	N
20	Wing & car damage	61	0.514	0.245	0.299	0.811	0.540	0.718	0.772	5
21	Braking & lockups	128	0.414	0.329	0.345	0.000	0.465	0.180	0.297	4
22	Wheel-to-wheel	71	0.325	0.440	0.517	0.959	0.696	0.211	0.717	6
23	Circuit action	72	0.033	0.021	0.010	0.761	0.222	0.236	0.136	1
24	Inter-car distance	60	0.025	0.197	0.023	0.833	0.230	0.145	0.137	1
25	Driver & team references	69	0.205	0.051	0.088	0.000	0.212	0.226	0.303	1
26	Defending & attacking	110	0.050	0.266	0.433	0.382	0.341	0.738	0.567	5
27	Overtake battles	762	0.412	0.177	0.493	0.054	0.543	0.299	0.398	4
28	Corner battles	82	0.222	0.230	0.258	0.929	0.364	0.378	0.373	4
29	Corner specific moves	60	0.141	0.174	0.192	0.822	0.267	0.276	0.273	1
30	Lap pace & timing	136	0.069	0.157	0.184	0.000	0.667	0.743	0.586	3
31	Race leader changes	162	0.088	0.181	0.019	0.364	0.658	0.554	0.304	4
32	Front position changes	45	0.105	0.093	0.106	0.000	0.109	0.153	0.174	0
33	Field position changes	122	0.280	0.245	0.261	0.918	0.352	0.364	0.430	4

S4 — Per-Cluster Four-Axis Validation Scores

This supplement tabulates the raw axis scores and binary pass/fail outcomes for all 34 clusters across the four validation axes. Thresholds: A1 (Algorithm) survival fraction $\geq 4/7 \approx 0.571$; A2 (Bootstrap) mean Jaccard ≥ 0.50 ; A3 (Held-out) replication rate ≥ 0.50 ; A4 (Hierarchical) mean Jaccard ≥ 0.50 . Clusters are sorted by verdict band then by size within band.

Table 2: Per-cluster raw scores and pass/fail indicators for all four validation axes. $\checkmark$ = criterion met; $\times$ = criterion not met. A1 score: survival fraction (0–1) across 7 pipelines. A2 score: mean Jaccard across 30 bootstrap subsamples. A3 score: held-out replication rate (rep_strict). A4 score: mean hierarchical Ward Jaccard.

C#	Label	Sz	A1	P	A2	P	A3	P	A4	P	Met	Band
6	Grand Prix wins	465	0.571	$\checkmark$	0.852	$\checkmark$	0.500	$\checkmark$	0.686	$\checkmark$	4	Strict-core
0	DRS-assisted battles	205	1.000	$\checkmark$	0.967	$\checkmark$	0.500	$\checkmark$	0.820	$\checkmark$	4	Strict-core
31	Race leader changes	162	0.571	$\checkmark$	0.635	$\checkmark$	0.500	$\checkmark$	0.606	$\checkmark$	4	Strict-core
9	Formula 1 milestones	134	0.571	$\checkmark$	0.797	$\checkmark$	0.846	$\checkmark$	0.671	$\checkmark$	4	Strict-core
21	Braking & lockups	128	0.571	$\checkmark$	0.579	$\checkmark$	0.600	$\checkmark$	0.532	$\checkmark$	4	Strict-core
16	Safety car deployment	122	1.000	$\checkmark$	0.689	$\checkmark$	0.562	$\checkmark$	0.626	$\checkmark$	4	Strict-core
1	Time penalties	118	1.000	$\checkmark$	0.966	$\checkmark$	0.926	$\checkmark$	0.798	$\checkmark$	4	Strict-core
8	Race control flags	118	0.857	$\checkmark$	0.772	$\checkmark$	0.605	$\checkmark$	0.729	$\checkmark$	4	Strict-core
28	Corner battles	82	0.571	$\checkmark$	0.617	$\checkmark$	0.500	$\checkmark$	0.675	$\checkmark$	4	Strict-core
22	Wheel-to-wheel	71	0.857	$\checkmark$	0.711	$\checkmark$	0.556	$\checkmark$	0.639	$\checkmark$	4	Strict-core
10	Chicane moves	62	0.857	$\checkmark$	0.853	$\checkmark$	1.000	$\checkmark$	0.823	$\checkmark$	4	Strict-core
20	Wing & car damage	61	0.714	$\checkmark$	0.626	$\checkmark$	0.625	$\checkmark$	0.670	$\checkmark$	4	Strict-core
11	Team radio celebrations	55	0.714	$\checkmark$	0.640	$\checkmark$	0.700	$\checkmark$	0.769	$\checkmark$	4	Strict-core
2	Lights-out race starts	54	1.000	$\checkmark$	0.876	$\checkmark$	0.600	$\checkmark$	0.703	$\checkmark$	4	Strict-core
3	Mirror checks defending	47	1.000	$\checkmark$	0.532	$\checkmark$	0.562	$\checkmark$	0.728	$\checkmark$	4	Strict-core
27	Overtake battles	762	0.571	$\checkmark$	0.688	$\checkmark$	0.556	$\checkmark$	0.339	$\times$	3	Robust
18	Team dynamics	600	0.857	$\checkmark$	0.709	$\checkmark$	1.000	$\checkmark$	0.365	$\times$	3	Robust
15	Tires & compounds	379	0.857	$\checkmark$	0.827	$\checkmark$	0.385	$\times$	0.575	$\checkmark$	3	Robust
14	Pit stops	185	1.000	$\checkmark$	0.776	$\checkmark$	0.667	$\checkmark$	0.447	$\times$	3	Robust
13	Championship & title	158	0.714	$\checkmark$	0.693	$\checkmark$	0.600	$\checkmark$	0.492	$\times$	3	Robust
12	Podium finishes	139	1.000	$\checkmark$	0.826	$\checkmark$	0.462	$\times$	0.767	$\checkmark$	3	Robust
30	Lap pace & timing	136	0.429	$\times$	0.697	$\checkmark$	0.900	$\checkmark$	0.534	$\checkmark$	3	Robust
33	Field position changes	122	0.571	$\checkmark$	0.510	$\checkmark$	0.619	$\checkmark$	0.392	$\times$	3	Robust
19	Race lead holding	115	0.857	$\checkmark$	0.694	$\checkmark$	0.300	$\times$	0.620	$\checkmark$	3	Robust
5	Weather conditions	43	1.000	$\checkmark$	0.075	$\times$	0.667	$\checkmark$	0.716	$\checkmark$	3	Robust
4	Race start launch	42	1.000	$\checkmark$	0.358	$\times$	0.600	$\checkmark$	0.861	$\checkmark$	3	Robust
26	Defending & attacking	110	0.714	$\checkmark$	0.664	$\checkmark$	0.333	$\times$	0.481	$\times$	2	Moderate
23	Circuit action	72	0.143	$\times$	0.700	$\checkmark$	0.400	$\times$	0.585	$\checkmark$	2	Moderate
17	Gap & pace	67	1.000	$\checkmark$	0.461	$\times$	0.333	$\times$	0.602	$\checkmark$	2	Moderate
7	Race start lights	36	0.429	$\times$	0.041	$\times$	0.636	$\checkmark$	0.864	$\checkmark$	2	Moderate
29	Corner specific moves	60	0.143	$\times$	0.445	$\times$	0.375	$\times$	0.717	$\checkmark$	1	Weak
25	Driver & team references	69	0.143	$\times$	0.472	$\times$	0.455	$\times$	0.481	$\times$	0	Fragile
24	Inter-car distance	60	0.143	$\times$	0.411	$\times$	0.417	$\times$	0.475	$\times$	0	Fragile
32	Front position changes	45	0.000	$\times$	0.241	$\times$		$\times$	0.407	$\times$	0	Fragile

S5 — Full R3 Enriched Vocabulary (864 FDR-Significant Terms)

This supplement lists all FDR-significant enriched terms for each of the 34 clusters, obtained by an exact one-sided hypergeometric test on binary sentence-level term incidence with a single global Benjamini–Hochberg family ($q \leq 0.05$). Terms are sorted by significance (descending) within each cluster. Over a vocabulary of 3,864 terms and 131,376 candidate (cluster, term) pairs, 864 are FDR-significant across 34 clusters (mean 25.4 per cluster, range 4–115; 54.6% bigrams; no cluster has an empty signature). The mask placeholders are stripped from the text before vectorisation, so <PERSON>, <TEAM> and <PLACE> cannot enter the lexicon while ordinary uses of *person*, *team* and *place* are retained. Columns: **Bi** = bigram indicator; z = significance rank score ($-\log_{10} p$); $\log_2 \text{FC}$ = $\log_2(\text{observed} / \text{expected frequency})$; n = observed count in cluster; **Cov%** = % of cluster sentences containing the term; **Corp%** = expected % under the hypergeometric null (expected / cluster size).

C0 — DRS-assisted battles

Domain: Race dynamics **Verdict:** Strict-core **Enriched terms:** 49

C0 — DRS-assisted battles: 49 FDR-significant terms

Term	Bi	z -score	$\log_2 \text{FC}$	n	Cov%	Corp%
drs		300.00	5.01	202	98.5%	3.1%
got drs	✓	50.80	5.10	34	16.6%	0.5%
drs zone	✓	22.60	5.13	15	7.3%	0.2%
drs got	✓	17.90	5.10	12	5.9%	0.2%
got		15.24	1.67	57	27.8%	8.7%
zone		14.19	3.64	17	8.3%	0.7%
open		12.27	3.74	14	6.8%	0.5%
drs enabled	✓	11.05	5.22	7	3.4%	0.1%
enabled		10.15	5.03	7	3.4%	0.1%
going drs	✓	10.15	5.03	7	3.4%	0.1%
bit drs	✓	9.46	5.22	6	2.9%	0.1%
drs going	✓	9.46	5.22	6	2.9%	0.1%
extra		8.88	3.74	10	4.9%	0.4%
drs open	✓	8.03	4.81	6	2.9%	0.1%
drs time	✓	7.88	5.22	5	2.4%	0.1%
ll drs	✓	7.88	5.22	5	2.4%	0.1%
second drs	✓	7.88	5.22	5	2.4%	0.1%
slipstream drs	✓	7.88	5.22	5	2.4%	0.1%
speed		7.84	2.62	15	7.3%	1.2%
straight		7.25	2.20	18	8.8%	1.9%
helping		5.82	4.37	5	2.4%	0.1%
rear wing	✓	5.66	3.81	6	2.9%	0.2%
200 mph	✓	5.48	3.72	6	2.9%	0.2%
help		5.27	4.08	5	2.4%	0.1%
defensive		5.15	2.94	8	3.9%	0.5%
extra pace	✓	5.14	4.63	4	2.0%	0.1%
slipstream		5.09	2.36	11	5.4%	1.0%
200		5.01	3.48	6	2.9%	0.3%
ll		4.87	1.79	16	7.8%	2.2%
drs getting	✓	4.72	5.23	3	1.5%	0.0%
drs goes	✓	4.72	5.23	3	1.5%	0.0%

Term	Bi	z-score	log ₂ FC	n	Cov%	Corp%
drs ll	✓	4.72	5.23	3	1.5%	0.0%
drs main	✓	4.72	5.23	3	1.5%	0.0%
drs range	✓	4.72	5.23	3	1.5%	0.0%
drs right	✓	4.72	5.23	3	1.5%	0.0%
drs turn	✓	4.72	5.23	3	1.5%	0.0%
line drs	✓	4.72	5.23	3	1.5%	0.0%
right drs	✓	4.72	5.23	3	1.5%	0.0%
straight drs	✓	4.72	5.23	3	1.5%	0.0%
wing open	✓	4.72	5.23	3	1.5%	0.0%
main straight	✓	4.67	2.98	7	3.4%	0.4%
mph		4.51	3.22	6	2.9%	0.3%
drs wide	✓	4.13	4.81	3	1.5%	0.1%
speed advantage	✓	4.13	4.81	3	1.5%	0.1%
speed drs	✓	4.13	4.81	3	1.5%	0.1%
chance		3.97	1.97	11	5.4%	1.4%
main		3.96	2.64	7	3.4%	0.5%
range		3.74	4.48	3	1.5%	0.1%
advantage		3.64	2.27	8	3.9%	0.8%

C1 — Time penalties

Domain: Regulatory **Verdict:** Strict-core **Enriched terms:** 39

C1 — Time penalties: 39 FDR-significant terms

Term	Bi	z-score	log ₂ FC	n	Cov%	Corp%
penalty		221.37	5.96	107	90.7%	1.5%
time penalty	✓	80.02	5.98	43	36.4%	0.6%
second penalty	✓	72.10	5.98	39	33.1%	0.5%
second		67.33	3.25	83	70.3%	7.4%
second time	✓	61.04	5.40	41	34.7%	0.8%
10 second	✓	47.70	5.92	27	22.9%	0.4%
causing collision	✓	35.08	6.02	19	16.1%	0.2%
penalty causing	✓	35.08	6.02	19	16.1%	0.2%
collision		34.77	5.77	21	17.8%	0.3%
causing		33.73	5.82	20	16.9%	0.3%
10		28.16	4.21	28	23.7%	1.3%
time		23.71	2.59	45	38.1%	6.3%
penalty forcing	✓	16.44	6.02	9	7.6%	0.1%
got second	✓	15.31	5.53	10	8.5%	0.2%
forcing		15.30	5.23	11	9.3%	0.2%
given second	✓	14.60	6.02	8	6.8%	0.1%
track		14.03	2.96	23	19.5%	2.5%
unsafe release	✓	12.76	6.02	7	5.9%	0.1%
given		12.63	4.10	13	11.0%	0.6%
gaining advantage	✓	12.40	5.56	8	6.8%	0.1%

Term	Bi	<i>z</i> -score	log ₂ FC	<i>n</i>	Cov%	Corp%
leaving track	✓	12.40	5.56	8	6.8%	0.1%
track gaining	✓	12.40	5.56	8	6.8%	0.1%
release		11.92	5.43	8	6.8%	0.2%
unsafe		11.21	5.65	7	5.9%	0.1%
leaving		11.15	5.21	8	6.8%	0.2%
penalty leaving	✓	10.92	6.01	6	5.1%	0.1%
forcing driver	✓	10.09	5.80	6	5.1%	0.1%
driver track	✓	9.09	6.02	5	4.2%	0.1%
given 10	✓	9.09	6.02	5	4.2%	0.1%
penalty speeding	✓	9.09	6.02	5	4.2%	0.1%
speeding pit	✓	9.09	6.02	5	4.2%	0.1%
advantage		8.71	3.52	11	9.3%	0.8%
speeding		7.78	5.53	5	4.2%	0.1%
gaining		7.59	3.97	8	6.8%	0.4%
got penalty	✓	6.57	5.70	4	3.4%	0.1%
penalty hanging	✓	5.45	6.03	3	2.5%	0.0%
penalty unsafe	✓	5.45	6.03	3	2.5%	0.0%
risk penalty	✓	5.45	6.03	3	2.5%	0.0%
gets second	✓	4.16	5.01	3	2.5%	0.1%

C2 — Lights-out race starts

Domain: Race dynamics **Verdict:** Strict-core **Enriched terms:** 16

C2 — Lights-out race starts: 16 FDR-significant terms

Term	Bi	<i>z</i> -score	log ₂ FC	<i>n</i>	Cov%	Corp%
lights away	✓	84.41	6.70	41	75.9%	0.7%
lights		78.89	5.92	45	83.3%	1.4%
away		55.42	4.90	41	75.9%	2.5%
away gets	✓	12.19	6.94	6	11.1%	0.1%
away getting	✓	7.96	6.84	4	7.4%	0.1%
start		6.86	2.93	11	20.4%	2.7%
lights lights	✓	6.48	7.16	3	5.6%	0.0%
away ahead	✓	5.88	6.74	3	5.6%	0.1%
start good	✓	5.88	6.74	3	5.6%	0.1%
getting away	✓	4.94	5.94	3	5.6%	0.1%
better start	✓	4.74	5.74	3	5.6%	0.1%
gets away	✓	3.94	4.92	3	5.6%	0.2%
away comes	✓	3.84	6.57	2	3.7%	0.0%
reaction		3.69	3.75	4	7.4%	0.5%
better		3.68	3.18	5	9.3%	1.0%
gets better	✓	3.54	6.16	2	3.7%	0.1%

C3 — Mirror checks defending

Domain: Race dynamics **Verdict:** Strict-core **Enriched terms:** 23

C3 — Mirror checks defending: 23 FDR-significant terms

Term		Bi z-score	log ₂ FC	n	Cov%	Corp%
mirrors		47.84	7.16	22	46.8%	0.3%
mirror		36.31	6.93	18	38.3%	0.3%
checking		21.24	6.81	11	23.4%	0.2%
scarlet		16.96	6.71	9	19.1%	0.2%
checks		16.31	7.04	8	17.0%	0.1%
checking mirrors	✓	13.41	7.34	6	12.8%	0.1%
checking left	✓	10.38	7.08	5	10.6%	0.1%
hand mirror	✓	10.38	7.08	5	10.6%	0.1%
left mirror	✓	8.90	7.32	4	8.5%	0.1%
looking mirrors	✓	8.90	7.32	4	8.5%	0.1%
left		8.30	3.62	10	21.3%	1.7%
checks mirrors	✓	6.66	7.38	3	6.4%	0.0%
looking		6.21	3.42	8	17.0%	1.6%
checks mirror	✓	6.06	6.91	3	6.4%	0.1%
driving mirrors	✓	6.06	6.91	3	6.4%	0.1%
left hand	✓	5.58	4.46	5	10.6%	0.5%
sees		5.07	4.89	4	8.5%	0.3%
hand		4.39	3.67	5	10.6%	0.8%
check		4.03	5.03	3	6.4%	0.2%
doesn need	✓	3.96	6.80	2	4.3%	0.0%
having look	✓	3.66	6.32	2	4.3%	0.1%
rival title	✓	3.66	6.32	2	4.3%	0.1%
trying retake	✓	3.66	6.32	2	4.3%	0.1%

C4 — Race start launch

Domain: Race dynamics **Verdict:** Robust **Enriched terms:** 15

C4 — Race start launch: 15 FDR-significant terms

Term		Bi z-score	log ₂ FC	n	Cov%	Corp%
launch		82.05	6.87	38	90.5%	0.8%
launch line	✓	41.10	7.24	19	45.2%	0.3%
good launch	✓	26.91	7.13	13	31.0%	0.2%
line		19.95	3.60	23	54.8%	4.5%
good		12.77	3.67	15	35.7%	2.8%
got good	✓	6.60	6.06	4	9.5%	0.1%
got launch	✓	6.21	7.09	3	7.1%	0.1%
gets launch	✓	5.82	6.80	3	7.1%	0.1%
chops		4.90	5.94	3	7.1%	0.1%
pole sitter	✓	4.90	5.94	3	7.1%	0.1%
sitter		4.90	5.94	3	7.1%	0.1%
chop		4.74	5.77	3	7.1%	0.1%
got		4.44	1.83	13	31.0%	8.7%
initial launch	✓	4.06	6.97	2	4.8%	0.0%

Term	Bi	<i>z</i> -score	log ₂ FC	<i>n</i>	Cov%	Corp%
great launch	✓	3.76	6.51	2	4.8%	0.1%

C5 — Weather conditions

Domain: Strategy & technical **Verdict:** Robust **Enriched terms:** 15

C5 — Weather conditions: 15 FDR-significant terms

Term	Bi	<i>z</i> -score	log ₂ FC	<i>n</i>	Cov%	Corp%
rain		69.90	7.08	32	74.4%	0.5%
rain coming	✓	8.36	7.16	4	9.3%	0.1%
wet		7.73	5.80	5	11.6%	0.2%
minutes		6.97	6.29	4	9.3%	0.1%
rain falling	✓	6.78	7.46	3	7.0%	0.0%
falling		5.24	6.27	3	7.0%	0.1%
light		4.34	5.36	3	7.0%	0.2%
expected		4.24	5.25	3	7.0%	0.2%
track		4.07	2.70	7	16.3%	2.5%
extraordinary		4.05	5.06	3	7.0%	0.2%
class		4.03	6.88	2	4.7%	0.0%
lasting		4.03	6.88	2	4.7%	0.0%
temperature		3.82	4.81	3	7.0%	0.2%
10 minutes	✓	3.74	6.51	2	4.7%	0.1%
30		3.51	6.16	2	4.7%	0.1%

C6 — Grand Prix wins

Domain: Narrative & meta **Verdict:** Strict-core **Enriched terms:** 115

C6 — Grand Prix wins: 115 FDR-significant terms

Term	Bi	<i>z</i> -score	log ₂ FC	<i>n</i>	Cov%	Corp%
grand		300.00	3.83	445	95.7%	6.7%
grand prix	✓	300.00	3.83	445	95.7%	6.7%
prix		300.00	3.82	454	97.6%	6.9%
wins		80.58	3.56	88	18.9%	1.6%
wins grand	✓	36.66	3.95	32	6.9%	0.4%
lead grand	✓	32.95	3.99	28	6.0%	0.4%
british		22.18	3.90	20	4.3%	0.3%
british grand	✓	22.18	3.90	20	4.3%	0.3%
win		17.24	2.22	42	9.0%	1.9%
place grand	✓	17.15	3.95	15	3.2%	0.2%
dutch		14.65	4.04	12	2.6%	0.2%
dutch grand	✓	14.65	4.04	12	2.6%	0.2%
prix victory	✓	13.43	4.04	11	2.4%	0.1%
lead		12.89	1.53	57	12.3%	4.3%
prix underway	✓	12.41	3.13	17	3.7%	0.4%

Term	Bi	z-score	$\log_2$ FC	n	Cov%	Corp%
australian		12.37	3.91	11	2.4%	0.2%
australian grand	✓	12.37	3.91	11	2.4%	0.2%
saudi		12.37	3.91	11	2.4%	0.2%
win grand	✓	12.20	4.04	10	2.2%	0.1%
belgian		11.58	3.80	11	2.4%	0.2%
belgian grand	✓	11.58	3.80	11	2.4%	0.2%
italian		11.58	3.80	11	2.4%	0.2%
italian grand	✓	11.58	3.80	11	2.4%	0.2%
lap grand	✓	10.98	4.04	9	1.9%	0.1%
victory		10.82	2.27	25	5.4%	1.1%
grands		10.43	3.78	10	2.2%	0.2%
grands prix	✓	10.43	3.78	10	2.2%	0.2%
underway		10.08	2.65	18	3.9%	0.6%
winner		9.81	2.70	17	3.7%	0.6%
arabian		9.75	4.04	8	1.7%	0.1%
arabian grand	✓	9.75	4.04	8	1.7%	0.1%
prix winner	✓	9.75	4.04	8	1.7%	0.1%
saudi arabian	✓	9.75	4.04	8	1.7%	0.1%
takes		9.64	1.78	33	7.1%	2.1%
victories		9.48	3.41	11	2.4%	0.2%
prix win	✓	8.82	3.87	8	1.7%	0.1%
canadian		8.15	3.72	8	1.7%	0.1%
canadian grand	✓	8.15	3.72	8	1.7%	0.1%
takes lead	✓	8.07	2.68	14	3.0%	0.5%
azerbaijan		7.65	3.85	7	1.5%	0.1%
azerbaijan grand	✓	7.65	3.85	7	1.5%	0.1%
austrian		7.61	3.58	8	1.7%	0.1%
austrian grand	✓	7.61	3.58	8	1.7%	0.1%
start grand	✓	7.61	3.58	8	1.7%	0.1%
formula grand	✓	7.31	4.04	6	1.3%	0.1%
started grand	✓	7.31	4.04	6	1.3%	0.1%
hungarian		7.02	3.68	7	1.5%	0.1%
hungarian grand	✓	7.02	3.68	7	1.5%	0.1%
japanese		7.02	3.68	7	1.5%	0.1%
japanese grand	✓	7.02	3.68	7	1.5%	0.1%
spanish		6.76	3.34	8	1.7%	0.2%
spanish grand	✓	6.76	3.34	8	1.7%	0.2%
row		6.51	2.32	14	3.0%	0.6%
prix going	✓	6.49	3.82	6	1.3%	0.1%
prix year	✓	6.11	3.39	7	1.5%	0.1%
leads grand	✓	6.09	4.04	5	1.1%	0.1%
prix comes	✓	6.09	4.04	5	1.1%	0.1%
end grand	✓	5.45	3.46	6	1.3%	0.1%
abu		5.33	3.78	5	1.1%	0.1%
abu dhabi	✓	5.33	3.78	5	1.1%	0.1%
dhabi		5.33	3.78	5	1.1%	0.1%

Term	Bi	z-score	$\log_2$ FC	n	Cov%	Corp%
dhabi grand	✓	5.33	3.78	5	1.1%	0.1%
prix second	✓	5.33	3.78	5	1.1%	0.1%
prix victories	✓	5.33	3.78	5	1.1%	0.1%
time wins	✓	5.33	3.78	5	1.1%	0.1%
year		5.27	1.62	20	4.3%	1.4%
emilia		4.87	4.04	4	0.9%	0.1%
emilia romagna	✓	4.87	4.04	4	0.9%	0.1%
lead british	✓	4.87	4.04	4	0.9%	0.1%
line wins	✓	4.87	4.04	4	0.9%	0.1%
position grand	✓	4.87	4.04	4	0.9%	0.1%
prix grand	✓	4.87	4.04	4	0.9%	0.1%
prix time	✓	4.87	4.04	4	0.9%	0.1%
retirement grand	✓	4.87	4.04	4	0.9%	0.1%
romagna		4.87	4.04	4	0.9%	0.1%
romagna grand	✓	4.87	4.04	4	0.9%	0.1%
wins abu	✓	4.87	4.04	4	0.9%	0.1%
wins australian	✓	4.87	4.04	4	0.9%	0.1%
wins austrian	✓	4.87	4.04	4	0.9%	0.1%
wins azerbaijan	✓	4.87	4.04	4	0.9%	0.1%
wins belgian	✓	4.87	4.04	4	0.9%	0.1%
wins british	✓	4.87	4.04	4	0.9%	0.1%
won grand	✓	4.87	4.04	4	0.9%	0.1%
second grand	✓	4.81	3.55	5	1.1%	0.1%
ends		4.41	3.36	5	1.1%	0.1%
started		4.31	1.83	13	2.8%	0.8%
chinese		4.19	3.72	4	0.9%	0.1%
chinese grand	✓	4.19	3.72	4	0.9%	0.1%
moment grand	✓	4.19	3.72	4	0.9%	0.1%
prix lights	✓	4.19	3.72	4	0.9%	0.1%
prix row	✓	4.19	3.72	4	0.9%	0.1%
victories row	✓	4.19	3.72	4	0.9%	0.1%
grid		4.16	1.79	13	2.8%	0.8%
final		4.11	1.45	18	3.9%	1.4%
home		4.05	1.55	16	3.4%	1.2%
winning		3.82	2.72	6	1.3%	0.2%
10th grid	✓	3.65	4.04	3	0.6%	0.0%
fifth grand	✓	3.65	4.04	3	0.6%	0.0%
final grand	✓	3.65	4.04	3	0.6%	0.0%
lead saudi	✓	3.65	4.04	3	0.6%	0.0%
leading grand	✓	3.65	4.04	3	0.6%	0.0%
meters grand	✓	3.65	4.04	3	0.6%	0.0%
prix got	✓	3.65	4.04	3	0.6%	0.0%
saudi grand	✓	3.65	4.04	3	0.6%	0.0%
seventh grand	✓	3.65	4.04	3	0.6%	0.0%
stop grand	✓	3.65	4.04	3	0.6%	0.0%
winner australian	✓	3.65	4.04	3	0.6%	0.0%

Term	Bi	z-score	log ₂ FC	n	Cov%	Corp%
wins canadian	✓	3.65	4.04	3	0.6%	0.0%
wins dutch	✓	3.65	4.04	3	0.6%	0.0%
wins emilia	✓	3.65	4.04	3	0.6%	0.0%
wins italian	✓	3.65	4.04	3	0.6%	0.0%
wins japanese	✓	3.65	4.04	3	0.6%	0.0%
opening		3.64	1.80	11	2.4%	0.7%
retirement		3.63	2.62	6	1.3%	0.2%
leads		3.61	1.42	16	3.4%	1.3%

C7 — Race start lights

Domain: Race dynamics **Verdict:** Moderate **Enriched terms:** 26

C7 — Race start lights: 26 FDR-significant terms

Term	Bi	z-score	log ₂ FC	n	Cov%	Corp%
lights		63.11	6.10	34	94.4%	1.4%
red lights	✓	27.71	7.62	12	33.3%	0.2%
prix underway	✓	27.65	6.63	15	41.7%	0.4%
underway		26.80	6.18	16	44.4%	0.6%
grand		21.04	3.36	25	69.4%	6.7%
grand prix	✓	21.04	3.36	25	69.4%	6.7%
prix		20.81	3.33	25	69.4%	6.9%
red		19.42	6.07	12	33.3%	0.5%
lights time	✓	18.03	7.57	8	22.2%	0.1%
lights away	✓	13.38	5.24	10	27.8%	0.7%
green flag	✓	10.77	5.84	7	19.4%	0.3%
begins		9.67	6.90	5	13.9%	0.1%
racing lights	✓	9.38	7.72	4	11.1%	0.1%
green		8.75	4.95	7	19.4%	0.6%
away		7.90	3.45	10	27.8%	2.5%
come lights	✓	7.54	6.72	4	11.1%	0.1%
away formula	✓	7.02	7.74	3	8.3%	0.0%
begins lights	✓	7.02	7.74	3	8.3%	0.0%
flag		6.89	3.69	8	22.2%	1.7%
racing		6.60	3.56	8	22.2%	1.9%
time grand	✓	5.72	6.74	3	8.3%	0.1%
time		4.27	2.13	10	27.8%	6.3%
begins red	✓	4.19	7.16	2	5.6%	0.0%
lights grand	✓	4.19	7.16	2	5.6%	0.0%
flag come	✓	3.89	6.72	2	5.6%	0.1%
formula goes	✓	3.89	6.72	2	5.6%	0.1%

C8 — Race control flags

Domain: Regulatory **Verdict:** Strict-core **Enriched terms:** 34

C8 — Race control flags: 34 FDR-significant terms

Term	Bi	<i>z</i> -score	log ₂ FC	<i>n</i>	Cov%	Corp%
flag		178.58	5.62	100	84.7%	1.7%
checkered		42.45	5.72	26	22.0%	0.4%
checkered flag	✓	42.45	5.72	26	22.0%	0.4%
yellow		38.17	5.65	24	20.3%	0.4%
green		33.09	5.08	25	21.2%	0.6%
yellow flag	✓	30.09	5.80	18	15.3%	0.3%
chequered		26.37	5.92	15	12.7%	0.2%
chequered flag	✓	26.37	5.92	15	12.7%	0.2%
green flag	✓	24.87	5.41	17	14.4%	0.3%
red flag	✓	13.37	5.09	10	8.5%	0.2%
red		12.70	4.36	12	10.2%	0.5%
flag flies	✓	10.92	6.01	6	5.1%	0.1%
sees checkered	✓	10.92	6.01	6	5.1%	0.1%
sees		10.81	4.73	9	7.6%	0.3%
nation driver	✓	10.09	3.49	13	11.0%	1.0%
flag sector	✓	10.09	5.80	6	5.1%	0.1%
flies		10.09	5.80	6	5.1%	0.1%
got yellow	✓	10.09	5.80	6	5.1%	0.1%
flags		8.62	5.28	6	5.1%	0.1%
sector		8.22	4.21	8	6.8%	0.4%
nation		8.15	2.84	14	11.9%	1.7%
black		7.27	6.01	4	3.4%	0.1%
flag fly	✓	7.27	6.01	4	3.4%	0.1%
driver		6.24	2.23	15	12.7%	2.7%
fly		6.10	5.43	4	3.4%	0.1%
driver checkered	✓	5.45	6.03	3	2.5%	0.0%
driver look	✓	5.45	6.03	3	2.5%	0.0%
flag flying	✓	5.45	6.03	3	2.5%	0.0%
flag wins	✓	5.45	6.03	3	2.5%	0.0%
sees chequered	✓	5.45	6.03	3	2.5%	0.0%
racing		4.91	2.31	11	9.3%	1.9%
going green	✓	4.85	5.60	3	2.5%	0.1%
waved		4.85	5.60	3	2.5%	0.1%
yellow flags	✓	4.46	5.28	3	2.5%	0.1%

C9 — Formula 1 milestones

Domain: Narrative & meta **Verdict:** Strict-core **Enriched terms:** 36

C9 — Formula 1 milestones: 36 FDR-significant terms

Term	Bi	<i>z</i> -score	log ₂ FC	<i>n</i>	Cov%	Corp%
formula		227.58	5.38	128	95.5%	2.3%
formula world	✓	21.08	5.63	13	9.7%	0.2%
time formula	✓	16.24	5.10	12	9.0%	0.3%

Term	Bi	<i>z</i> -score	log ₂ FC	<i>n</i>	Cov%	Corp%
podium formula	✓	14.20	5.54	9	6.7%	0.1%
podium		14.13	2.89	24	17.9%	2.4%
formula career	✓	11.47	5.64	7	5.2%	0.1%
world championship	✓	10.79	3.47	14	10.4%	0.9%
world		10.31	2.82	18	13.4%	1.9%
championship		9.15	2.60	18	13.4%	2.2%
career		8.37	3.92	9	6.7%	0.4%
best formula	✓	7.04	5.84	4	3.0%	0.1%
formula 2025	✓	7.04	5.84	4	3.0%	0.1%
formula podium	✓	7.04	5.84	4	3.0%	0.1%
2025		6.75	4.42	6	4.5%	0.2%
time		6.67	1.62	26	19.4%	6.3%
champion		6.21	2.93	10	7.5%	1.0%
returns		5.88	5.25	4	3.0%	0.1%
personal		5.52	5.02	4	3.0%	0.1%
welcome		5.52	5.02	4	3.0%	0.1%
champion formula	✓	5.28	5.82	3	2.2%	0.0%
formula 2022	✓	5.28	5.82	3	2.2%	0.0%
formula champion	✓	5.28	5.82	3	2.2%	0.0%
formula returns	✓	5.28	5.82	3	2.2%	0.0%
welcome formula	✓	5.28	5.82	3	2.2%	0.0%
season		5.07	2.36	11	8.2%	1.6%
victory		4.80	2.60	9	6.7%	1.1%
win formula	✓	4.68	5.42	3	2.2%	0.1%
personal best	✓	4.29	5.09	3	2.2%	0.1%
sport		4.29	5.09	3	2.2%	0.1%
win		4.28	2.08	11	8.2%	1.9%
time year	✓	3.99	4.84	3	2.2%	0.1%
victory formula	✓	3.99	4.84	3	2.2%	0.1%
2023		3.76	4.61	3	2.2%	0.1%
stand		3.75	3.75	4	3.0%	0.2%
driving		3.74	3.20	5	3.7%	0.4%
best		3.57	2.45	7	5.2%	1.0%

C10 — Chicane moves

Domain: Race dynamics **Verdict:** Strict-core **Enriched terms:** 8

C10 — Chicane moves: 8 FDR-significant terms

Term	Bi	<i>z</i> -score	log ₂ FC	<i>n</i>	Cov%	Corp%
chicane		123.42	6.35	60	96.8%	1.2%
second chicane	✓	13.24	6.58	7	11.3%	0.1%
going chicane	✓	11.81	6.72	6	9.7%	0.1%
goes chicane	✓	6.29	6.97	3	4.8%	0.0%
braking		4.26	3.14	6	9.7%	1.1%
bold		3.97	4.95	3	4.8%	0.2%

Term	Bi	z -score	$\log_2$ FC	n	Cov%	Corp%
chicane going	✓	3.71	6.38	2	3.2%	0.0%
make chicane	✓	3.71	6.38	2	3.2%	0.0%

C11 — Team radio celebrations

Domain: Narrative & meta **Verdict:** Strict-core **Enriched terms:** 36

C11 — Team radio celebrations: 36 FDR-significant terms

Term	Bi	z -score	$\log_2$ FC	n	Cov%	Corp%
thank		34.93	6.09	21	38.2%	0.6%
job		21.25	5.76	14	25.5%	0.5%
mate		20.25	5.87	13	23.6%	0.4%
weekend		19.30	5.09	15	27.3%	0.8%
guys		19.03	5.31	14	25.5%	0.6%
amazing		11.55	5.53	8	14.5%	0.3%
fantastic		11.38	6.02	7	12.7%	0.2%
thanks		10.50	5.68	7	12.7%	0.2%
unbelievable		10.50	5.68	7	12.7%	0.2%
great job	✓	7.45	6.54	4	7.3%	0.1%
thank guys	✓	7.45	6.54	4	7.3%	0.1%
drive		7.40	3.93	8	14.5%	1.0%
good job	✓	7.08	6.32	4	7.3%	0.1%
yeah		6.91	2.60	13	23.6%	3.9%
awesome		6.45	7.09	3	5.5%	0.0%
deserve		6.45	7.09	3	5.5%	0.0%
job guys	✓	6.45	7.09	3	5.5%	0.0%
job mate	✓	6.45	7.09	3	5.5%	0.0%
weekend thank	✓	6.45	7.09	3	5.5%	0.0%
great		6.45	3.23	9	16.4%	1.7%
love		6.12	4.80	5	9.1%	0.3%
nice		5.83	4.03	6	10.9%	0.7%
good		5.62	2.69	10	18.2%	2.8%
world champions	✓	4.92	5.91	3	5.5%	0.1%
yes		4.74	3.42	6	10.9%	1.0%
difficult		4.64	4.53	4	7.3%	0.3%
beautiful		4.39	5.38	3	5.5%	0.1%
really really	✓	4.39	5.38	3	5.5%	0.1%
champions		4.02	5.00	3	5.5%	0.2%
year		3.97	2.96	6	10.9%	1.4%
really good	✓	3.92	4.89	3	5.5%	0.2%
really		3.87	2.60	7	12.7%	2.1%
congrats		3.82	6.51	2	3.6%	0.0%
executed		3.82	6.51	2	3.6%	0.0%
good weekend	✓	3.82	6.51	2	3.6%	0.0%
yes yes	✓	3.82	6.51	2	3.6%	0.0%

C12 — Podium finishes**Domain:** Narrative & meta **Verdict:** Robust **Enriched terms:** 36

C12 — Podium finishes: 36 FDR-significant terms

Term	Bi	<i>z</i> -score	log ₂ FC	<i>n</i>	Cov%	Corp%
podium		213.72	5.24	127	91.4%	2.4%
podium places	✓	21.20	5.42	14	10.1%	0.2%
home		18.75	3.75	22	15.8%	1.2%
place podium	✓	13.06	5.61	8	5.8%	0.1%
comes home	✓	10.55	4.30	10	7.2%	0.4%
podium place	✓	10.49	5.78	6	4.3%	0.1%
completes		10.25	4.87	8	5.8%	0.2%
places		10.00	3.13	15	10.8%	1.2%
finish		9.94	3.64	12	8.6%	0.7%
completes podium	✓	9.76	5.13	7	5.0%	0.1%
podium finish	✓	9.06	5.37	6	4.3%	0.1%
complete podium	✓	7.96	5.52	5	3.6%	0.1%
step		7.63	4.34	7	5.0%	0.2%
step podium	✓	7.56	4.78	6	4.3%	0.2%
stand podium	✓	7.42	5.30	5	3.6%	0.1%
come home	✓	7.30	4.67	6	4.3%	0.2%
podium season	✓	6.29	5.46	4	2.9%	0.1%
rostrum		6.29	5.46	4	2.9%	0.1%
complete		5.72	3.91	6	4.3%	0.3%
career podium	✓	5.23	5.77	3	2.2%	0.0%
double podium	✓	5.23	5.77	3	2.2%	0.0%
final podium	✓	5.23	5.77	3	2.2%	0.0%
gonna podium	✓	5.23	5.77	3	2.2%	0.0%
seventh podium	✓	5.23	5.77	3	2.2%	0.0%
home second	✓	5.16	4.10	5	3.6%	0.2%
stand		5.02	4.02	5	3.6%	0.2%
season		4.92	2.31	11	7.9%	1.6%
time		4.75	1.39	23	16.5%	6.3%
home podium	✓	4.63	5.36	3	2.2%	0.1%
career		4.54	3.28	6	4.3%	0.4%
campaign		3.95	4.78	3	2.2%	0.1%
finish season	✓	3.95	4.78	3	2.2%	0.1%
final		3.85	2.20	9	6.5%	1.4%
home finish	✓	3.71	4.56	3	2.2%	0.1%
return		3.51	4.37	3	2.2%	0.1%
place		3.51	1.09	24	17.3%	8.1%

C13 — Championship & title**Domain:** Narrative & meta **Verdict:** Robust **Enriched terms:** 36

C13 — Championship & title: 36 FDR-significant terms

Term	Bi	z -score	$\log_2$ FC	n	Cov%	Corp%
championship		84.59	4.40	74	46.8%	2.2%
world		78.05	4.48	67	42.4%	1.9%
world championship	✓	36.87	4.47	33	20.9%	0.9%
champion		36.12	4.41	33	20.9%	1.0%
world champion	✓	19.44	4.23	19	12.0%	0.6%
title		14.78	4.11	15	9.5%	0.5%
constructors		13.78	4.54	12	7.6%	0.3%
win		13.68	2.91	23	14.6%	1.9%
time world	✓	11.78	4.60	10	6.3%	0.3%
championship leader	✓	11.25	4.46	10	6.3%	0.3%
trophy		9.79	5.08	7	4.4%	0.1%
2025		9.51	4.60	8	5.1%	0.2%
points		8.23	2.72	15	9.5%	1.4%
leader		7.69	3.20	11	7.0%	0.8%
2024		7.52	4.72	6	3.8%	0.1%
seven time	✓	7.23	4.60	6	3.8%	0.2%
champions		6.97	4.48	6	3.8%	0.2%
time champion	✓	6.76	5.59	4	2.5%	0.1%
world title	✓	6.76	5.59	4	2.5%	0.1%
championship battle	✓	6.38	4.75	5	3.2%	0.1%
champion world	✓	6.06	5.28	4	2.5%	0.1%
championship lead	✓	5.83	4.46	5	3.2%	0.1%
constructors championship	✓	5.60	4.33	5	3.2%	0.2%
constructors title	✓	5.06	5.60	3	1.9%	0.0%
gap world	✓	5.06	5.60	3	1.9%	0.0%
2022		5.05	4.01	5	3.2%	0.2%
seven		4.48	2.91	7	4.4%	0.6%
constructors champions	✓	4.47	5.18	3	1.9%	0.1%
tally		4.47	5.18	3	1.9%	0.1%
win world	✓	4.47	5.18	3	1.9%	0.1%
time		4.30	1.26	24	15.2%	6.3%
today		4.29	2.57	8	5.1%	0.9%
championship points	✓	4.08	4.86	3	1.9%	0.1%
score		3.96	3.89	4	2.5%	0.2%
contenders		3.70	3.69	4	2.5%	0.2%
winner		3.64	2.76	6	3.8%	0.6%

C14 — Pit stops

Domain: Strategy & technical **Verdict:** Robust **Enriched terms:** 41

C14 — Pit stops: 41 FDR-significant terms

Term	Bi	z -score	$\log_2$ FC	n	Cov%	Corp%
pit		188.89	4.61	139	75.1%	3.1%
pit stop	✓	58.45	4.52	50	27.0%	1.2%
lane		58.41	4.70	47	25.4%	1.0%
pit lane	✓	57.04	4.75	45	24.3%	0.9%
stop		47.71	3.36	63	34.1%	3.3%
pits		38.81	3.97	41	22.2%	1.4%
coming pit	✓	11.36	5.37	7	3.8%	0.1%
comes pit	✓	8.30	4.95	6	3.2%	0.1%
come pits	✓	6.56	4.25	6	3.2%	0.2%
comes		6.47	1.73	23	12.4%	3.7%
coming		6.28	1.99	18	9.7%	2.4%
come pit	✓	5.79	5.05	4	2.2%	0.1%
pitted		5.59	3.42	7	3.8%	0.4%
stops		5.48	3.37	7	3.8%	0.4%
slow		5.39	2.63	10	5.4%	0.9%
fast lane	✓	5.32	4.79	4	2.2%	0.1%
second pit	✓	4.96	4.56	4	2.2%	0.1%
come		4.93	1.74	17	9.2%	2.7%
coming pits	✓	4.88	3.88	5	2.7%	0.2%
60 kilometers	✓	4.86	5.36	3	1.6%	0.0%
high pressure	✓	4.86	5.36	3	1.6%	0.0%
pit crew	✓	4.86	5.36	3	1.6%	0.0%
slow pit	✓	4.86	5.36	3	1.6%	0.0%
started pit	✓	4.86	5.36	3	1.6%	0.0%
starting pit	✓	4.86	5.36	3	1.6%	0.0%
time pit	✓	4.86	5.36	3	1.6%	0.0%
make pit	✓	4.67	4.37	4	2.2%	0.1%
60		4.26	4.95	3	1.6%	0.1%
called pits	✓	4.26	4.95	3	1.6%	0.1%
exit pit	✓	4.26	4.95	3	1.6%	0.1%
pits comes	✓	4.26	4.95	3	1.6%	0.1%
starting		3.93	3.30	5	2.7%	0.3%
cheap		3.87	4.63	3	1.6%	0.1%
crews		3.87	4.63	3	1.6%	0.1%
called		3.85	3.79	4	2.2%	0.2%
crew		3.85	3.79	4	2.2%	0.2%
pit stops	✓	3.85	3.79	4	2.2%	0.2%
fast		3.64	3.11	5	2.7%	0.3%
going pit	✓	3.58	4.37	3	1.6%	0.1%
kilometers hour	✓	3.58	4.37	3	1.6%	0.1%
pits going	✓	3.58	4.37	3	1.6%	0.1%

C15 — Tires & compounds

Domain: Strategy & technical **Verdict:** Robust **Enriched terms:** 83

Term	Bi	z -score	$\log_2$ FC	n	Cov%	Corp%
tires		200.07	4.05	170	44.9%	2.7%
tire		106.09	4.01	96	25.3%	1.6%
tyres		67.12	4.13	58	15.3%	0.9%
hard		65.81	3.60	74	19.5%	1.6%
medium		51.97	3.97	49	12.9%	0.8%
compound		36.33	4.16	31	8.2%	0.5%
tyre		33.46	4.07	30	7.9%	0.5%
soft		31.02	3.86	31	8.2%	0.6%
fresher		26.27	3.80	27	7.1%	0.5%
compound tire	✓	22.33	4.33	17	4.5%	0.2%
fresher tires	✓	20.59	4.11	18	4.7%	0.3%
hard compound	✓	19.80	4.25	16	4.2%	0.2%
medium tires	✓	18.87	4.16	16	4.2%	0.2%
hard tires	✓	16.93	3.84	17	4.5%	0.3%
intermediate		16.80	3.94	16	4.2%	0.3%
temperature		16.18	3.99	15	4.0%	0.2%
hard tyres	✓	15.73	4.33	12	3.2%	0.2%
set		15.09	3.37	19	5.0%	0.5%
hard tire	✓	15.07	3.85	15	4.0%	0.3%
medium tire	✓	15.07	4.13	13	3.4%	0.2%
tires going	✓	12.56	4.09	11	2.9%	0.2%
soft tire	✓	12.08	4.20	10	2.6%	0.1%
soft tires	✓	11.36	3.89	11	2.9%	0.2%
intermediate tires	✓	11.32	4.07	10	2.6%	0.2%
rubber		10.80	3.15	15	4.0%	0.4%
hard tyre	✓	10.47	4.34	8	2.1%	0.1%
fresh		9.72	3.75	10	2.6%	0.2%
slick		9.72	3.75	10	2.6%	0.2%
traction		9.25	2.64	17	4.5%	0.7%
medium compound	✓	9.16	4.33	7	1.8%	0.1%
rear		8.46	2.25	20	5.3%	1.1%
new		8.17	2.63	15	4.0%	0.6%
medium tyre	✓	7.85	4.34	6	1.6%	0.1%
set tires	✓	7.85	4.34	6	1.6%	0.1%
compound tires	✓	7.02	4.11	6	1.6%	0.1%
fired		7.02	4.11	6	1.6%	0.1%
grip		6.69	2.21	16	4.2%	0.9%
pits		6.59	1.90	20	5.3%	1.4%
new set	✓	6.44	3.92	6	1.6%	0.1%
pirelli		6.44	3.92	6	1.6%	0.1%
set hard	✓	5.78	4.07	5	1.3%	0.1%
softs		5.78	4.07	5	1.3%	0.1%
better		5.30	1.96	15	4.0%	1.0%
cold		5.25	3.85	5	1.3%	0.1%

Term	Bi	z-score	log ₂ FC	n	Cov%	Corp%
warm		5.25	3.85	5	1.3%	0.1%
12 laps	✓	5.23	4.34	4	1.1%	0.1%
compound tyre	✓	5.23	4.34	4	1.1%	0.1%
new tires	✓	5.23	4.34	4	1.1%	0.1%
rear axle	✓	5.23	4.34	4	1.1%	0.1%
tire temperature	✓	5.23	4.34	4	1.1%	0.1%
tires got	✓	5.23	4.34	4	1.1%	0.1%
tires lot	✓	5.23	4.34	4	1.1%	0.1%
younger		5.23	4.34	4	1.1%	0.1%
rid		4.99	3.33	6	1.6%	0.2%
window		4.74	3.22	6	1.6%	0.2%
axle		4.54	4.01	4	1.1%	0.1%
try end	✓	4.54	4.01	4	1.1%	0.1%
end		4.31	1.52	18	4.7%	1.7%
harder		4.08	3.75	4	1.1%	0.1%
soft compound	✓	4.08	3.75	4	1.1%	0.1%
tires just	✓	4.08	3.75	4	1.1%	0.1%
working		4.07	2.62	7	1.8%	0.3%
bring		3.94	2.56	7	1.8%	0.3%
bring tires	✓	3.92	4.33	3	0.8%	0.0%
fresh tires	✓	3.92	4.33	3	0.8%	0.0%
got tyres	✓	3.92	4.33	3	0.8%	0.0%
intermediate tyres	✓	3.92	4.33	3	0.8%	0.0%
laps younger	✓	3.92	4.33	3	0.8%	0.0%
temperature window	✓	3.92	4.33	3	0.8%	0.0%
tire grip	✓	3.92	4.33	3	0.8%	0.0%
tire medium	✓	3.92	4.33	3	0.8%	0.0%
tire run	✓	3.92	4.33	3	0.8%	0.0%
tire way	✓	3.92	4.33	3	0.8%	0.0%
tires gets	✓	3.92	4.33	3	0.8%	0.0%
tires getting	✓	3.92	4.33	3	0.8%	0.0%
tires make	✓	3.92	4.33	3	0.8%	0.0%
tires ve	✓	3.92	4.33	3	0.8%	0.0%
tyres fired	✓	3.92	4.33	3	0.8%	0.0%
tyres got	✓	3.92	4.33	3	0.8%	0.0%
warming		3.92	4.33	3	0.8%	0.0%
faster		3.76	3.07	5	1.3%	0.2%
fresher rubber	✓	3.76	3.07	5	1.3%	0.2%
mediums		3.57	2.96	5	1.3%	0.2%

C16 — Safety car deployment

Domain: Regulatory **Verdict:** Strict-core **Enriched terms:** 25

C16 — Safety car deployment: 25 FDR-significant terms

Term	Bi	<i>z</i> -score	log ₂ FC	<i>n</i>	Cov%	Corp%
safety		190.33	5.59	106	86.9%	1.8%
safety car	✓	189.72	5.62	105	86.1%	1.8%
car		129.58	4.09	109	89.3%	5.2%
virtual		58.43	5.81	34	27.9%	0.5%
virtual safety	✓	56.55	5.81	33	27.0%	0.5%
deployed		22.03	5.61	14	11.5%	0.2%
car deployed	✓	17.42	5.62	11	9.0%	0.2%
car safety	✓	12.65	5.97	7	5.7%	0.1%
ending		10.16	5.32	7	5.7%	0.1%
car ending	✓	9.02	5.97	5	4.1%	0.1%
going safety	✓	7.21	5.97	4	3.3%	0.1%
got safety	✓	7.21	5.97	4	3.3%	0.1%
lapped cars	✓	7.21	5.97	4	3.3%	0.1%
timing		6.65	4.97	5	4.1%	0.1%
car coming	✓	6.52	5.64	4	3.3%	0.1%
lapped		6.52	5.64	4	3.3%	0.1%
allowed overtake	✓	5.40	5.97	3	2.5%	0.0%
need safety	✓	5.40	5.97	3	2.5%	0.0%
barrier		5.03	3.55	6	4.9%	0.4%
car ll	✓	4.81	5.55	3	2.5%	0.1%
impact		4.81	5.55	3	2.5%	0.1%
end lap	✓	4.26	4.16	4	3.3%	0.2%
cars		4.17	2.53	8	6.6%	1.1%
surprised		3.88	4.74	3	2.5%	0.1%
barriers		3.68	3.16	5	4.1%	0.5%

C17 — Gap & pace

Domain: Strategy & technical **Verdict:** Moderate **Enriched terms:** 20

C17 — Gap & pace: 20 FDR-significant terms

Term	Bi	<i>z</i> -score	log ₂ FC	<i>n</i>	Cov%	Corp%
seconds		41.63	4.76	34	50.7%	1.9%
gap		26.50	4.82	22	32.8%	1.2%
tenths		10.21	5.03	8	11.9%	0.4%
hour		9.69	5.32	7	10.4%	0.3%
miles hour	✓	8.26	5.98	5	7.5%	0.1%
tenths second	✓	8.05	5.18	6	9.0%	0.2%
gap seconds	✓	6.73	6.04	4	6.0%	0.1%
miles		6.60	5.07	5	7.5%	0.2%
half		5.59	3.90	6	9.0%	0.6%
extra		5.43	4.35	5	7.5%	0.4%
20		4.93	4.75	4	6.0%	0.2%

Term	Bi	<i>z</i> -score	log ₂ FC	<i>n</i>	Cov%	Corp%
pace		4.55	3.31	6	9.0%	0.9%
10 seconds	✓	4.46	5.42	3	4.5%	0.1%
second		4.07	1.60	15	22.4%	7.4%
10		3.70	2.80	6	9.0%	1.3%
12 miles	✓	3.65	6.27	2	3.0%	0.0%
gap come	✓	3.65	6.27	2	3.0%	0.0%
gap tenths	✓	3.65	6.27	2	3.0%	0.0%
second gap	✓	3.65	6.27	2	3.0%	0.0%
tenths quicker	✓	3.65	6.27	2	3.0%	0.0%

C18 — Team dynamics

Domain: Narrative & meta **Verdict:** Robust **Enriched terms:** 5

C18 — Team dynamics: 5 FDR-significant terms

Term	Bi	<i>z</i> -score	log ₂ FC	<i>n</i>	Cov%	Corp%
teammate		27.82	2.24	65	10.8%	2.3%
team		16.57	2.26	38	6.3%	1.3%
ahead teammate	✓	6.29	3.09	8	1.3%	0.2%
ahead		4.42	0.78	56	9.3%	5.5%
past		3.52	0.73	48	8.0%	4.8%

C19 — Race lead holding

Domain: Race dynamics **Verdict:** Robust **Enriched terms:** 14

C19 — Race lead holding: 14 FDR-significant terms

Term	Bi	<i>z</i> -score	log ₂ FC	<i>n</i>	Cov%	Corp%
lead		64.46	3.80	68	59.1%	4.3%
leads		46.95	4.73	39	33.9%	1.3%
leads second	✓	7.42	5.38	5	4.3%	0.1%
lead second	✓	7.31	6.06	4	3.5%	0.1%
maintains lead	✓	7.07	5.21	5	4.3%	0.1%
holds lead	✓	6.78	5.06	5	4.3%	0.1%
lead comes	✓	6.62	5.74	4	3.5%	0.1%
holds		6.21	4.18	6	5.2%	0.3%
leads comes	✓	5.78	5.25	4	3.5%	0.1%
maintains		5.57	4.37	5	4.3%	0.2%
leads inside	✓	5.48	6.06	3	2.6%	0.0%
leads leads	✓	4.88	5.64	3	2.6%	0.1%
takes lead	✓	4.86	3.47	6	5.2%	0.5%
maintain lead	✓	3.95	4.84	3	2.6%	0.1%

C20 — Wing & car damage

Domain: Strategy & technical **Verdict:** Strict-core **Enriched terms:** 16

C20 — Wing & car damage: 16 FDR-significant terms

Term	Bi	<i>z</i> -score	log ₂ FC	<i>n</i>	Cov%	Corp%
wing		64.35	6.21	36	59.0%	0.8%
puncture		24.18	6.26	14	23.0%	0.3%
damage		21.42	5.24	16	26.2%	0.7%
damaged		16.82	6.55	9	14.8%	0.2%
got puncture	✓	8.82	6.29	5	8.2%	0.1%
contact		8.80	3.33	12	19.7%	2.0%
bit contact	✓	8.44	6.97	4	6.6%	0.1%
damage wing	✓	8.44	6.97	4	6.6%	0.1%
damaged wing	✓	8.44	6.97	4	6.6%	0.1%
rear		6.52	3.56	8	13.1%	1.1%
wing endplate	✓	6.32	6.97	3	4.9%	0.0%
think		6.19	2.72	11	18.0%	2.7%
endplate		5.72	6.55	3	4.9%	0.1%
ripped		5.72	6.55	3	4.9%	0.1%
rear wing	✓	5.21	4.97	4	6.6%	0.2%
wing damage	✓	3.73	6.38	2	3.3%	0.0%

C21 — Braking & lockups

Domain: Race dynamics **Verdict:** Strict-core **Enriched terms:** 19

C21 — Braking & lockups: 19 FDR-significant terms

Term	Bi	<i>z</i> -score	log ₂ FC	<i>n</i>	Cov%	Corp%
braking		72.31	5.18	51	39.8%	1.1%
brakes		43.39	4.47	39	30.5%	1.4%
braking zone	✓	27.51	5.18	20	15.6%	0.4%
zone		22.49	4.55	20	15.6%	0.7%
brake		16.72	4.96	13	10.2%	0.3%
late		12.62	3.26	18	14.1%	1.5%
late braking	✓	10.01	5.25	7	5.5%	0.1%
brake pedal	✓	8.92	5.90	5	3.9%	0.1%
pedal		8.14	5.64	5	3.9%	0.1%
lock		5.45	3.09	8	6.2%	0.7%
brakes just	✓	5.34	5.91	3	2.3%	0.0%
way traffic	✓	5.34	5.91	3	2.3%	0.0%
outside brakes	✓	4.74	5.48	3	2.3%	0.1%
curb		4.49	3.70	5	3.9%	0.3%
brakes late	✓	4.35	5.16	3	2.3%	0.1%
late brakes	✓	4.14	3.47	5	3.9%	0.4%
rear		4.09	2.49	8	6.2%	1.1%
super late	✓	4.05	4.91	3	2.3%	0.1%
zone turn	✓	4.05	4.91	3	2.3%	0.1%

C22 — Wheel-to-wheel**Domain:** Race dynamics **Verdict:** Strict-core **Enriched terms:** 11

C22 — Wheel-to-wheel: 11 FDR-significant terms

Term	Bi	<i>z</i> -score	log ₂ FC	<i>n</i>	Cov%	Corp%
wheel		110.01	5.64	64	90.1%	1.8%
wheel wheel	✓	88.81	5.93	51	71.8%	1.2%
going wheel	✓	26.22	5.99	16	22.5%	0.4%
wheel turn	✓	10.85	6.34	6	8.5%	0.1%
wheel goes	✓	6.12	6.74	3	4.2%	0.0%
goes wheel	✓	5.52	6.34	3	4.2%	0.1%
went wheel	✓	5.52	6.34	3	4.2%	0.1%
wheel action	✓	5.52	6.34	3	4.2%	0.1%
wheel gets	✓	5.52	6.34	3	4.2%	0.1%
wheels		5.22	4.22	5	7.0%	0.4%
went		4.45	2.92	7	9.9%	1.3%

C23 — Circuit action**Domain:** Race dynamics **Verdict:** Moderate **Enriched terms:** 4

C23 — Circuit action: 4 FDR-significant terms

Term	Bi	<i>z</i> -score	log ₂ FC	<i>n</i>	Cov%	Corp%
racetrack		109.79	6.16	58	80.6%	1.1%
corner racetrack	✓	9.42	6.48	5	6.9%	0.1%
turn racetrack	✓	6.97	6.16	4	5.6%	0.1%
turns racetrack	✓	3.58	6.16	2	2.8%	0.0%

C24 — Inter-car distance**Domain:** Strategy & technical **Verdict:** Fragile **Enriched terms:** 17

C24 — Inter-car distance: 17 FDR-significant terms

Term	Bi	<i>z</i> -score	log ₂ FC	<i>n</i>	Cov%	Corp%
car		40.14	3.74	42	70.0%	5.2%
cars		13.15	4.25	13	21.7%	1.1%
car length	✓	11.90	6.77	6	10.0%	0.1%
car lengths	✓	11.30	6.57	6	10.0%	0.1%
length		11.30	6.57	6	10.0%	0.1%
lengths		11.30	6.57	6	10.0%	0.1%
couple car	✓	7.77	6.68	4	6.7%	0.1%
width		7.29	6.41	4	6.7%	0.1%
got car	✓	6.38	5.82	4	6.7%	0.1%
length ahead	✓	6.34	6.97	3	5.0%	0.0%
car ahead	✓	5.79	5.41	4	6.7%	0.2%

Term	Bi	<i>z</i> -score	log ₂ FC	<i>n</i>	Cov%	Corp%
car width	✓	5.34	6.27	3	5.0%	0.1%
flat floor	✓	5.04	6.00	3	5.0%	0.1%
floor		4.14	5.12	3	5.0%	0.1%
flat		4.02	5.00	3	5.0%	0.2%
couple		3.90	3.34	5	8.3%	0.8%
positions car	✓	3.74	6.38	2	3.3%	0.0%

C25 — Driver & team references

Domain: Narrative & meta **Verdict:** Fragile **Enriched terms:** 8

C25 — Driver & team references: 8 FDR-significant terms

Term	Bi	<i>z</i> -score	log ₂ FC	<i>n</i>	Cov%	Corp%
driver		62.54	4.71	49	71.0%	2.7%
nation driver	✓	37.60	5.32	27	39.1%	1.0%
nation		32.21	4.61	28	40.6%	1.7%
drivers		13.68	4.38	13	18.8%	0.9%
driver getting	✓	6.15	6.80	3	4.3%	0.0%
board nation	✓	5.55	6.38	3	4.3%	0.1%
driver trying	✓	5.55	6.38	3	4.3%	0.1%
contract		3.62	6.21	2	2.9%	0.0%

C26 — Defending & attacking

Domain: Race dynamics **Verdict:** Moderate **Enriched terms:** 11

C26 — Defending & attacking: 11 FDR-significant terms

Term	Bi	<i>z</i> -score	log ₂ FC	<i>n</i>	Cov%	Corp%
fight		30.05	4.19	30	27.3%	1.5%
defend		24.69	5.05	19	17.3%	0.5%
battle		16.43	3.66	20	18.2%	1.4%
defensive		11.41	4.29	11	10.0%	0.5%
going fight	✓	8.81	5.38	6	5.5%	0.1%
try defend	✓	4.94	5.69	3	2.7%	0.1%
defended		4.55	5.38	3	2.7%	0.1%
try fight	✓	4.31	4.21	4	3.6%	0.2%
defense		3.70	3.73	4	3.6%	0.3%
fight		3.61	3.66	4	3.6%	0.3%
going		3.52	1.02	26	23.6%	11.7%

C27 — Overtake battles

Domain: Race dynamics **Verdict:** Robust **Enriched terms:** 17

C27 — Overtake battles: 17 FDR-significant terms

Term	Bi	<i>z</i> -score	log ₂ FC	<i>n</i>	Cov%	Corp%
inside		26.71	1.43	126	16.5%	6.1%
door		10.04	2.23	22	2.9%	0.6%
corner		7.47	1.15	50	6.6%	3.0%
gets away	✓	7.20	2.84	10	1.3%	0.2%
past		6.92	0.91	69	9.1%	4.8%
outside		6.65	0.93	64	8.4%	4.4%
open door	✓	6.45	3.01	8	1.0%	0.1%
getaway		5.44	2.40	10	1.3%	0.2%
space		5.13	1.64	18	2.4%	0.8%
ahead		4.64	0.71	68	8.9%	5.5%
close		4.03	0.93	36	4.7%	2.5%
goes corner	✓	4.01	3.33	4	0.5%	0.1%
leads corner	✓	4.01	3.33	4	0.5%	0.1%
past outside	✓	4.01	3.33	4	0.5%	0.1%
goes		3.92	0.64	67	8.8%	5.6%
inside goes	✓	3.53	2.23	7	0.9%	0.2%
inside line	✓	3.50	1.19	20	2.6%	1.2%

C28 — Corner battles

Domain: Race dynamics **Verdict:** Strict-core **Enriched terms:** 7

C28 — Corner battles: 7 FDR-significant terms

Term	Bi	<i>z</i> -score	log ₂ FC	<i>n</i>	Cov%	Corp%
turn		68.84	3.53	71	86.6%	7.5%
run turn	✓	10.39	5.54	7	8.5%	0.2%
abreast		6.38	5.74	4	4.9%	0.1%
way turn	✓	5.65	4.47	5	6.1%	0.3%
run		4.86	2.84	8	9.8%	1.4%
heading turn	✓	4.81	4.63	4	4.9%	0.2%
outside turn	✓	3.96	3.96	4	4.9%	0.3%

C29 — Corner specific moves

Domain: Race dynamics **Verdict:** Weak **Enriched terms:** 18

C29 — Corner specific moves: 18 FDR-significant terms

Term	Bi	<i>z</i> -score	log ₂ FC	<i>n</i>	Cov%	Corp%
turn		56.34	3.61	55	91.7%	7.5%
turn 11	✓	33.82	6.76	17	28.3%	0.3%
11		28.82	6.18	17	28.3%	0.4%
turn 10	✓	13.34	6.62	7	11.7%	0.1%
turn seven	✓	11.30	6.57	6	10.0%	0.1%

Term	Bi	<i>z</i> -score	log ₂ FC	<i>n</i>	Cov%	Corp%
13		7.32	5.51	5	8.3%	0.2%
turn 12	✓	6.99	5.31	5	8.3%	0.2%
12		6.61	4.45	6	10.0%	0.5%
inside turn	✓	6.57	5.07	5	8.3%	0.2%
seven		5.94	4.09	6	10.0%	0.6%
heading turn	✓	5.36	5.08	4	6.7%	0.2%
heading		5.27	3.72	6	10.0%	0.8%
coming turn	✓	5.24	4.99	4	6.7%	0.2%
turn 13	✓	5.04	6.00	3	5.0%	0.1%
10		4.99	3.19	7	11.7%	1.3%
14		4.82	4.67	4	6.7%	0.3%
turn 14	✓	4.02	5.00	3	5.0%	0.2%
away turn	✓	3.74	6.38	2	3.3%	0.0%

C30 — Lap pace & timing

Domain: Strategy & technical **Verdict:** Robust **Enriched terms:** 23

C30 — Lap pace & timing: 23 FDR-significant terms

Term	Bi	<i>z</i> -score	log ₂ FC	<i>n</i>	Cov%	Corp%
lap		196.75	4.55	135	99.3%	4.2%
fastest lap	✓	31.10	4.84	25	18.4%	0.6%
fastest		26.22	4.33	25	18.4%	0.9%
opening		14.18	4.02	15	11.0%	0.7%
previous lap	✓	14.09	5.82	8	5.9%	0.1%
opening lap	✓	13.94	4.88	11	8.1%	0.3%
previous		11.89	5.35	8	5.9%	0.1%
lap lap	✓	9.71	5.60	6	4.4%	0.1%
turn lap	✓	7.47	5.33	5	3.7%	0.1%
lap goes	✓	5.85	5.22	4	2.9%	0.1%
lap going	✓	5.85	5.22	4	2.9%	0.1%
lap time	✓	5.85	5.22	4	2.9%	0.1%
quicker		5.65	3.87	6	4.4%	0.3%
end lap	✓	5.54	4.33	5	3.7%	0.2%
middle		4.67	3.35	6	4.4%	0.4%
corners lap	✓	4.66	5.40	3	2.2%	0.1%
sector lap	✓	4.66	5.40	3	2.2%	0.1%
seconds		4.36	2.11	11	8.1%	1.9%
middle sector	✓	4.22	4.11	4	2.9%	0.2%
lap number	✓	3.98	4.81	3	2.2%	0.1%
sector		3.93	3.33	5	3.7%	0.4%
point fastest	✓	3.74	4.60	3	2.2%	0.1%
seconds away	✓	3.54	4.40	3	2.2%	0.1%

C31 — Race leader changes**Domain:** Race dynamics **Verdict:** Strict-core **Enriched terms:** 18

C31 — Race leader changes: 18 FDR-significant terms

Term	Bi	<i>z</i> -score	log ₂ FC	<i>n</i>	Cov%	Corp%
race		192.43	3.96	157	96.9%	6.2%
lead race	✓	39.33	5.12	28	17.3%	0.5%
lead		21.59	2.61	42	25.9%	4.2%
race leader	✓	10.07	4.13	10	6.2%	0.4%
leader		9.83	3.40	13	8.0%	0.8%
takes lead	✓	7.40	3.56	9	5.6%	0.5%
race lead	✓	6.08	4.06	6	3.7%	0.2%
start race	✓	5.78	4.42	5	3.1%	0.1%
started race	✓	5.55	4.98	4	2.5%	0.1%
win race	✓	4.65	4.39	4	2.5%	0.1%
going lead	✓	4.43	4.24	4	2.5%	0.1%
race going	✓	4.43	4.24	4	2.5%	0.1%
race control	✓	4.24	4.10	4	2.5%	0.1%
lap race	✓	4.21	3.49	5	3.1%	0.3%
race far	✓	4.04	4.82	3	1.9%	0.1%
review		4.04	4.82	3	1.9%	0.1%
place race	✓	3.66	3.65	4	2.5%	0.2%
retake lead	✓	3.51	4.34	3	1.9%	0.1%

C32 — Front position changes**Domain:** Race dynamics **Verdict:** Fragile **Enriched terms:** 7

C32 — Front position changes: 7 FDR-significant terms

Term	Bi	<i>z</i> -score	log ₂ FC	<i>n</i>	Cov%	Corp%
fourth		14.30	4.02	15	33.3%	2.1%
place		13.36	2.65	23	51.1%	8.1%
position		8.73	2.75	15	33.3%	5.0%
fourth place	✓	7.84	4.54	7	15.6%	0.7%
second		4.85	1.96	13	28.9%	7.4%
second place	✓	4.53	2.94	7	15.6%	2.0%
couple positions	✓	3.70	6.38	2	4.4%	0.1%

C33 — Field position changes**Domain:** Race dynamics **Verdict:** Robust **Enriched terms:** 16

Term	Bi	z-score	$\log_2$ FC	n	Cov%	Corp%
seventh		31.45	4.66	27	22.1%	0.9%
place		27.81	2.47	55	45.1%	8.1%
sixth		24.11	4.22	24	19.7%	1.1%
fifth		16.02	3.73	19	15.6%	1.2%
sixth place	✓	14.90	4.85	12	9.8%	0.3%
eighth		14.32	3.88	16	13.1%	0.9%
seventh place	✓	12.65	4.90	10	8.2%	0.3%
11th		11.50	4.59	10	8.2%	0.3%
position		11.32	2.21	28	23.0%	5.0%
fifth place	✓	8.73	4.05	9	7.4%	0.4%
sixth position	✓	8.54	4.33	8	6.6%	0.3%
10th		7.97	4.11	8	6.6%	0.4%
ninth		5.66	3.19	8	6.6%	0.7%
12th		5.44	4.29	5	4.1%	0.2%
seventh position	✓	4.40	4.27	4	3.3%	0.2%
11th place	✓	3.51	4.38	3	2.5%	0.1%

S6 — Per-Race Full-Broadcast Coverage Analysis

This supplement provides the full per-race coverage statistics for the ten held-out full-race broadcast transcripts (16,876 sentences; races from 2023–2025). Table 1 reports coverage at the operational threshold $\tau = 0.40$. Table 2 reports coverage sensitivity across $\tau \in [0.30, 0.50]$, together with the number of leftover clusters and the count of clusters that appear in all ten races (universal leftover clusters). Of the 59 leftover clusters at $\tau = 0.40$, 26 appear in all ten races, against a race-label-shuffle null expectation of 37 ± 3 .

Table 37: Per-race full-broadcast coverage at $\tau = 0.40$. Covered = cosine ≥ 0.40 to nearest cluster centroid; Leftover = cosine < 0.40 .

Race	Sentences	Covered	Leftover	Coverage %	Median cosine
2023 Australia	2,466	1,777	689	72.1%	0.538
2023 Qatar	1,336	954	382	71.4%	0.526
2023 Singapore	1,635	1,313	322	80.3%	0.590
2024 Bahrain	1,322	949	373	71.8%	0.519
2024 Brazil	1,905	1,421	484	74.6%	0.542
2024 Italy	1,458	1,101	357	75.5%	0.558
2024 Lasvegas	1,344	1,033	311	76.9%	0.566
2024 Qatar	1,631	1,225	406	75.1%	0.545
2025 Belgium	2,232	1,612	620	72.2%	0.519
2025 Japan	1,547	1,175	372	76.0%	0.554
Total / Pooled	16,876	12,560	4,316	74.4%	—

Table 38: Coverage sensitivity across τ thresholds. Leftover n : uncovered sentences; LC: leftover clusters after re-clustering; Universal: clusters present in all 10 held-out races.

τ	Coverage %	Leftover n	LC total	Universal
0.3	88.8%	1895	8	8
0.35	81.3%	3164	15	11
0.4	74.4%	4316	22	17
0.45	67.2%	5534	21	19
0.5	58.5%	7001	36	26

S7 — Leftover Cluster Catalog (Full-Race Coverage Residual)

This supplement catalogs the leftover clusters identified in the full-race coverage residual (Section 5.4 of the main paper). Each leftover cluster represents a thematic register present in full-race broadcasts but absent or underrepresented in the highlights-derived taxonomy. Universal clusters (present in all held-out races) are marked with a checkmark. The 60 leftover clusters were obtained by re-clustering the

10-race leftover sentences (cosine < 0.40 to any primary cluster centroid) with HDBSCAN using the same hyperparameters as the primary pipeline. Of these, **27** are universal (present in all 10 held-out races). Under a race-label-shuffle null (1,000 permutations of race-session labels across the leftover sentences, leftover partition held fixed) the expected number of universal clusters is 37 ± 3 ; the observed count falls below every permutation drawn ($p < 0.001$), indicating session-specific register structure rather than uniform background noise.

Table 39: All 60 leftover clusters from the full-race coverage residual. Universal: present in all 10 held-out races. Top terms derived by word-frequency from entity-masked sentences.

LC#	Size	Races	Univ.	Top terms
LC0	22	8/10		rule, rules, book, rulebook, allowed, apply, going, record
LC1	23	7/10		sector, middle, sectors, moment, best, got, good, yet
LC2	21	8/10		call, big, bluff, yeah, bluffing, right, let, spade
LC3	36	7/10		commentary, box, louis, replay, time, thought, yeah, going
LC4	52	9/10		medium, soft, hards, hard, mediums, going, softs, set
LC5	65	10/10	✓	weekend, every, haven, going, earlier, week, yeah, far
LC6	32	8/10		grid, boxes, row, take, starts, second, man, today
LC7	44	10/10	✓	top, drive, science, great, drives, way, third, fourth
LC8	97	10/10	✓	radio, engineer, hear, yeah, message, got, going, say
LC9	28	8/10		order, end, establish, running, time, got, right, off
LC10	88	9/10		fans, crowd, always, great, cheer, yeah, hope, big
LC11	20	9/10		move, board, yeah, nice, make, moves, like, great
LC12	32	7/10		nation, time, yeah, going, cup, tea, way, want
LC13	23	8/10		lock, again, front, locking, easier, seen, lockups, locked
LC14	45	8/10		restart, going, order, think, see, procedure, again, reset
LC15	21	9/10		screen, change, red, led, bring, carefully, like, switch
LC16	25	10/10	✓	pass, slam, dunk, passes, going, clean, see, difficult
LC17	45	10/10	✓	field, flat, pole, whole, front, pretty, year, position
LC18	23	6/10		year, years, season, remember, like, form, case, since
LC19	22	9/10		pace, know, got, running, run, good, going, yeah
LC20	49	9/10		start, yeah, good, going, got, rolling, opportunity, best
LC21	22	9/10		result, expect, yeah, fault, end, expecting, expected, sure
LC22	39	10/10	✓	keep, head, lot, got, helmet, hat, sport, everyone
LC23	29	9/10		end, long, way, think, going, try, got, feel
LC24	60	10/10	✓	tough, conditions, difficult, going, hard, easy, know, bit
LC25	50	10/10	✓	season, long, good, year, enough, rookies, start, yeah
LC26	36	8/10		points, today, getting, way, marks, think, going, everything
LC27	43	10/10	✓	sargent, twenty, fight, started, start, birthday, weekend, starting
LC28	46	10/10	✓	seen, year, time, times, like, moment, haven, past
LC29	59	10/10	✓	grip, speed, low, fuel, bit, high, little, corners
LC30	50	9/10		circuit, around, corners, circuits, turn, part, time, area

continued. . .

LC#	Size	Races	Univ.	Top terms
LC31	20	8/10		push, pushing, going, lot, keep, tell, defend, soft
LC32	21	4/10		dry, flat, tarmac, drying, drier, yeah, bit, already
LC33	32	9/10		wide, line, bit, white, yeah, grass, see, lines
LC34	35	10/10	✓	corner, corners, inside, lot, got, make, wall, itself
LC35	33	10/10	✓	temperature, degrees, air, bit, heat, hot, feel, track
LC36	50	10/10	✓	haven, again, away, going, moment, gone, thrown, catching
LC37	25	8/10		gravel, debris, kerbs, like, bit, kerb, trap, surface
LC38	31	10/10	✓	front, left, yeah, side, looking, right, sides, dead
LC39	27	10/10	✓	cockpit, air, clean, little, bit, around, see, got
LC40	30	10/10	✓	teams, year, start, got, bit, something, difficult, haven
LC41	46	10/10	✓	strategy, stop, think, like, different, expecting, done, right
LC42	49	9/10		start, aborted, procedure, formation, right, lap, procedures, investigation
LC43	126	10/10	✓	know, going, risk, think, done, got, looking, need
LC44	24	9/10		inside, pack, air, keep, got, outside, getting, going
LC45	36	10/10	✓	feel, yeah, happy, feeling, right, unwell, like, poised
LC46	76	10/10	✓	going, interesting, think, chance, happen, anything, fascinating, doing
LC47	20	8/10		stint, time, little, think, bit, end, summer, like
LC48	367	10/10	✓	got, think, going, way, time, yeah, done, know
LC49	22	10/10	✓	hear, heard, talk, want, means, talked, love, times
LC50	26	9/10		fun, yeah, lot, stuff, like, ahead, apex, scared
LC51	26	9/10		control, better, bit, yeah, job, think, fell, like
LC52	32	8/10		day, drama, today, dramatic, going, mean, makes, yesterday
LC53	50	9/10		strange, incredible, good, quite, god, got, off, stuff
LC54	180	10/10	✓	yeah, think, know, thing, question, bit, look, little
LC55	60	10/10	✓	close, yeah, surprised, bit, thought, let, know, far
LC56	23	9/10		again, boys, honestly, earlier, sorry, yeah, further, think
LC57	38	10/10	✓	got, going, find, underway, see, good, wonder, let
LC58	59	9/10		let, see, look, yeah, wait, check, again, okay
LC-1	1505	10/10	✓	going, yeah, bit, got, little, think, know, like

S8 — Sensitivity Analysis: MCS Sweep and LOO Pipeline

This supplement reports two pre-specified sensitivity analyses. Section 1 shows the effect of varying `min_cluster_size` (`mcs`) from 15 to 120 on key clustering metrics (cluster count, noise rate, DBCV, silhouette, vocabulary richness); the `mcs` = 10 point cited in Section 4.2 is from a separate extension run. The chosen value `mcs` = 35 (highlighted) sits at the centre of a stability plateau. Section 2 reports leave-one-out sensitivity for the Algorithm axis: how much the 27/34 majority-threshold count shifts when each of the seven alternative pipelines is removed.

1. `min_cluster_size` sensitivity sweep

Table 40: Sensitivity of the clustering pipeline to `min_cluster_size` (`mcs`). The highlighted row (`mcs` = 35) is the value used in the main analysis. DBCV: Density-Based Cluster Validation (Moulávi et al., 2014). Distinct. terms: mean distinctive unigrams per cluster under the sweep’s screen (log-odds > 2.0 at frequency ≥ 3) – a cheap ranking statistic, *not* the FDR-corrected R3 count of Section 4.3 (25.4 at `mcs` = 35); the scales are not comparable. Interp: clusters with ≥ 5 such terms.

<code>mcs</code>	<i>n</i> clusters	Noise	DBCV	Silhouette	Distinct. terms	Interp.
15	62	35.2%	0.353	0.633	11.3	89%
20	44	30.4%	0.346	0.640	13.8	91%
25	38	29.4%	0.303	0.635	15.3	97%
30	35	34.4%	0.360	0.667	15.3	94%
35	34	33.5%	0.348	0.662	17.2	100%
40	33	34.2%	0.322	0.669	16.7	97%
45	30	34.6%	0.316	0.658	16.8	100%
50	29	35.7%	0.328	0.660	17.0	100%
55	3	0.0%	0.360	0.347	18.7	100%
60	3	0.0%	0.360	0.347	18.7	100%
70	3	0.0%	0.360	0.347	18.7	100%
80	3	0.0%	0.360	0.347	18.7	100%
100	3	0.0%	0.360	0.347	18.7	100%
120	2	0.0%	0.332	0.494	15.5	50%

2. Leave-one-out pipeline sensitivity

The full Algorithm-axis majority count (all 7 judges) is 27/34. The table below shows how this count changes when each pipeline is removed from the judge panel (6 remaining pipelines; majority threshold $\geq 4/6$). No single pipeline is disproportionately influential.

Table 41: Leave-one-out (LOO) sensitivity: Algorithm-axis majority-threshold count when each pipeline is removed in turn. Counts are noise-adjusted, matching the 27/34 Algorithm-axis figure reported in the main text; full count (all 7 judges) = 27/34, Δ = change from 27.

Removed pipeline	Majority count	Δ
OPTICS	22	-5
Ward	22	-5
KMeans	23	-4
A1_MPNNet	26	-1
A2_BGE	26	-1
A3_GTE	25	-2
B2_Leiden	25	-2